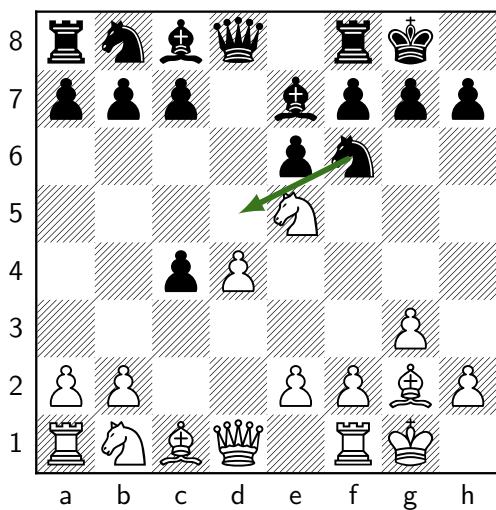


# Melanie-ANALOGY — Catalan

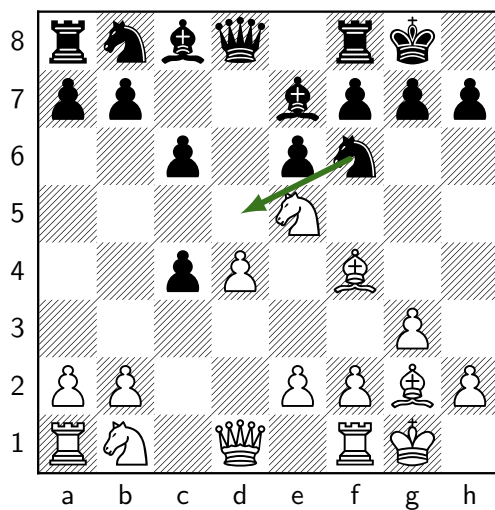
Melanie (Deep\_Vision)

## Abstract

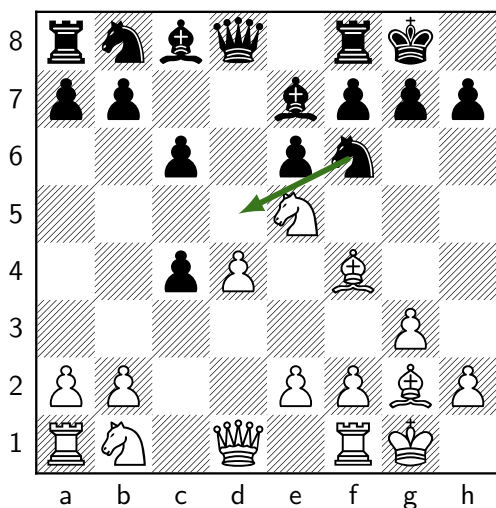
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under  $\mu$  – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



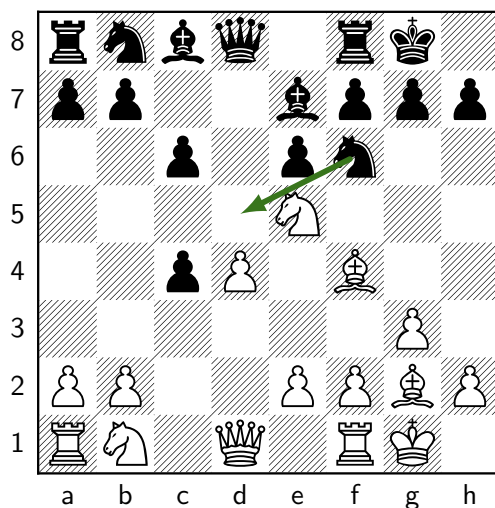
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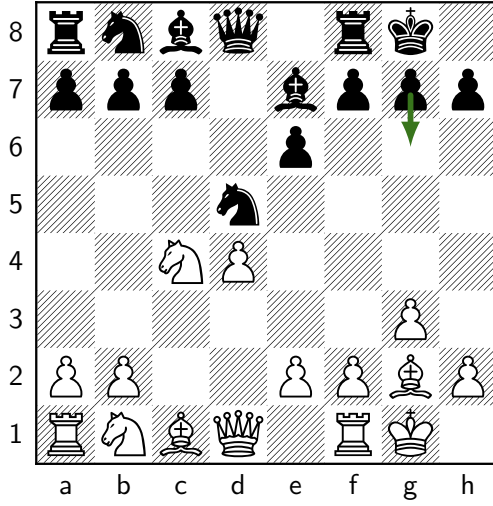


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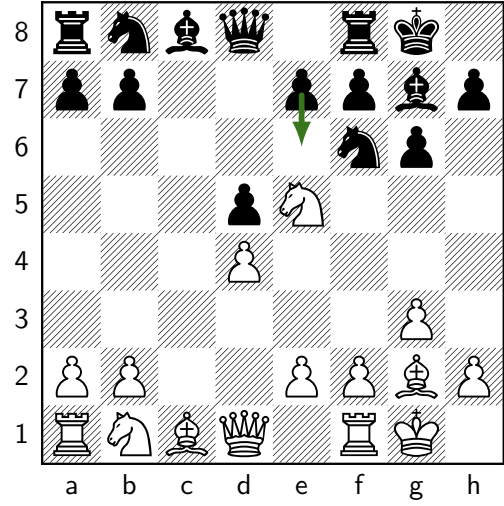


(d)

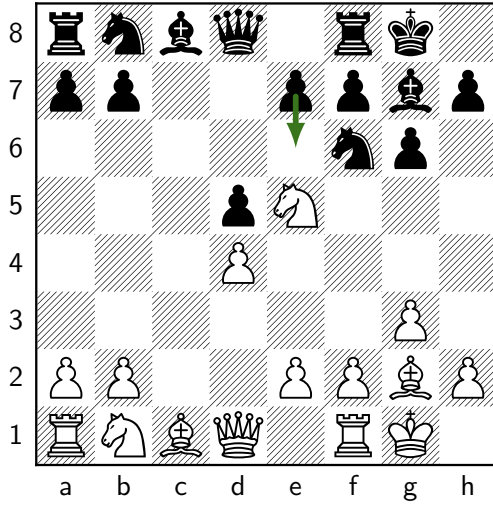
Figure 1: **Move 7, Black: Nd5.** (a) Our position; Melanie plays Nd5 (f6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#271` (ply 17) [structural distance  $d=26$ , QAP obj=146]: the master played Nd5; under  $\mu$  our  $Nf3$  plays the role of that game’s  $Nf3$ ; cross-type roles: our  $ne5$  as their  $bg2$ ; transferred move f6d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#272` (ply 17) [structural distance  $d=26$ , QAP obj=146]: the master played Nd5; under  $\mu$  our  $Nf3$  plays the role of that game’s  $Nf3$ ; cross-type roles: our  $ne5$  as their  $bg2$ ; transferred move f6d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3736` (ply 17) [structural distance  $d=26$ , QAP obj=146]: the master played Nd5; under  $\mu$  our  $Nf3$  plays the role of that game’s  $Nf3$ ; cross-type roles: our  $ne5$  as their  $bg2$ ; transferred move f6d5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nd5 (summed QAP objective 875, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



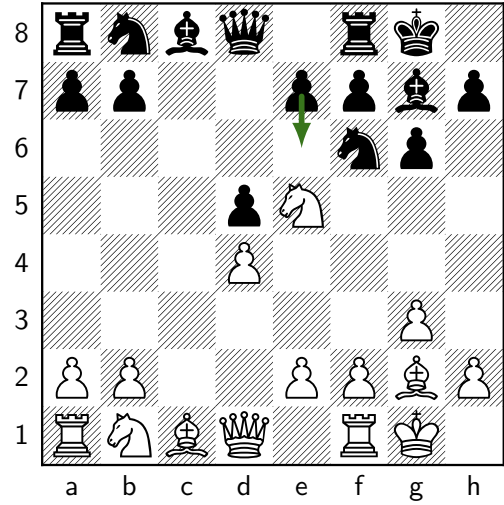
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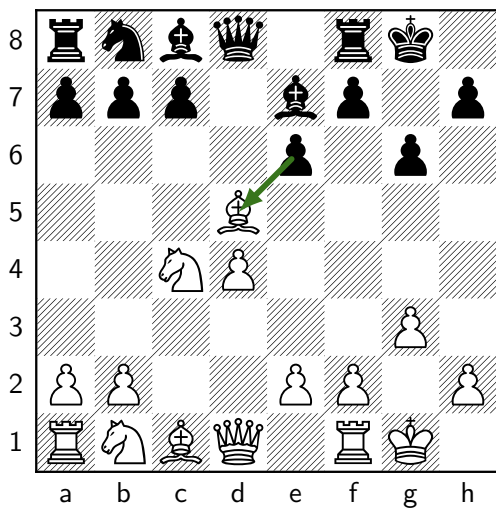


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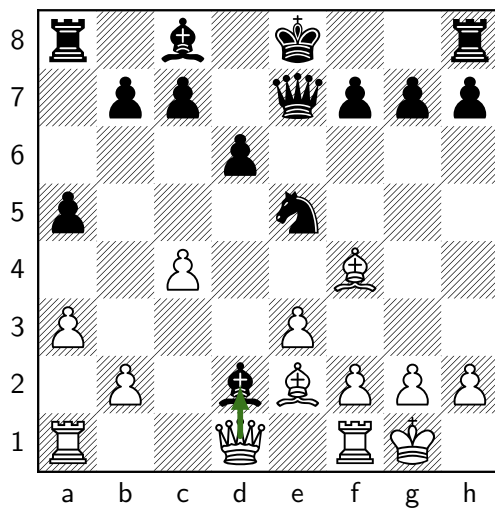


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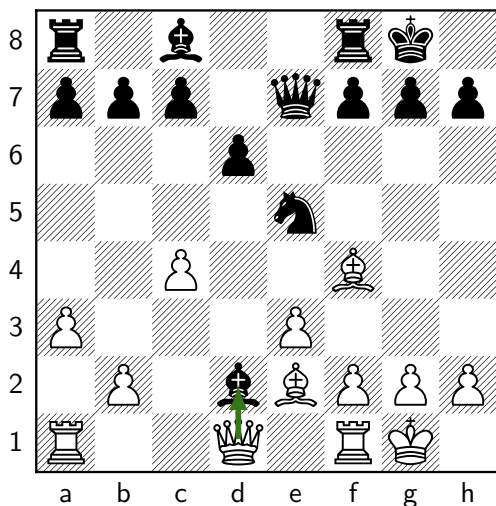
Figure 2: **Move 8, Black: g6.** (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0001#88` (ply 15) [structural distance  $d=48$ , QAP obj=20]: the master played e6; under  $\mu$  our  $Pg2$  plays the role of that game’s  $Pe2$ ; cross-type roles: our  $Be7$  as their  $Ra8$ , our  $Pb7$  as their  $Nb8$ ; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccr1_top.p0001#89` (ply 15) [structural distance  $d=48$ , QAP obj=20]: the master played e6; under  $\mu$  our  $Pg2$  plays the role of that game’s  $Pe2$ ; cross-type roles: our  $Be7$  as their  $Ra8$ , our  $Pb7$  as their  $Nb8$ ; transferred move g7g6 (**endorses this move**). (d) Retrieved from `ccr1_top.p0003#2420` (ply 15) [structural distance  $d=48$ , QAP obj=20]: the master played e6; under  $\mu$  our  $Pg2$  plays the role of that game’s  $Pe2$ ; cross-type roles: our  $Be7$  as their  $Ra8$ , our  $Pb7$  as their  $Nb8$ ; transferred move g7g6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to g6 (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



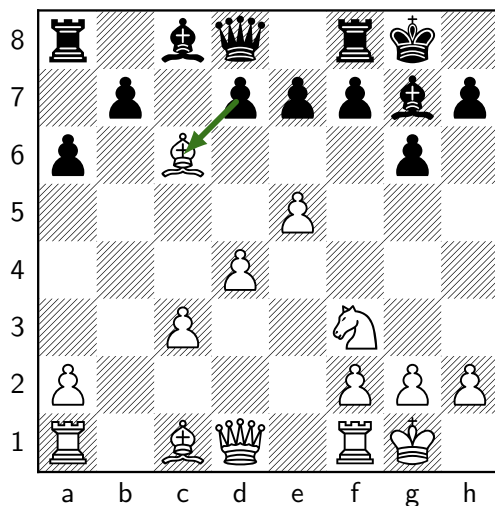
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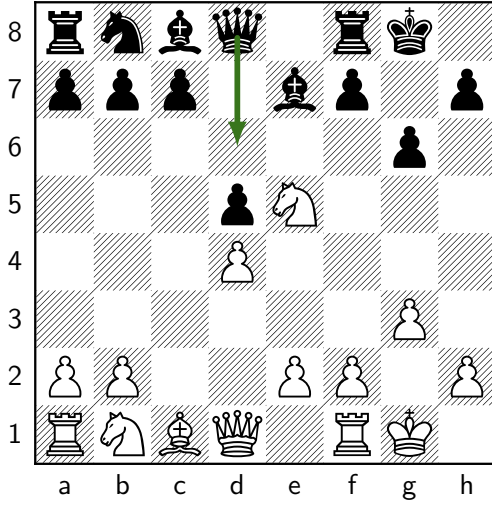


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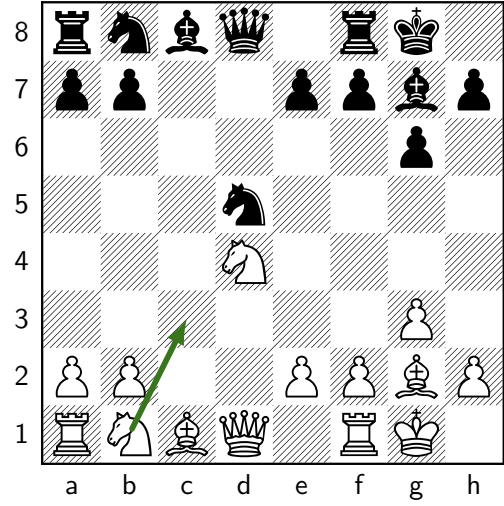


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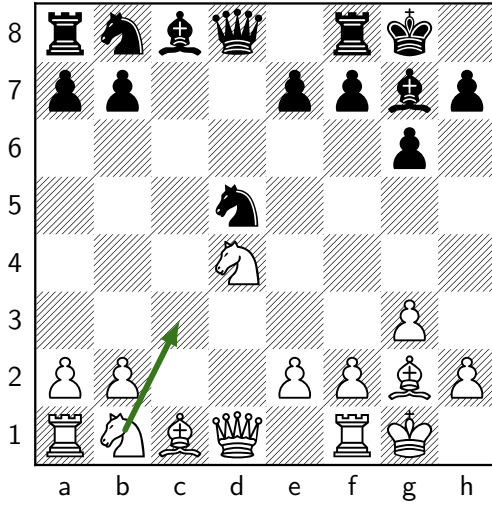
Figure 3: **Move 9, Black: exd5.** (a) Our position; Melanie plays **exd5** (**e6d5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1714` (ply 22) [structural distance  $d=51$ , QAP obj=62]: the master played **Qxd2**; under  $\mu$  our *Pe3* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Be7* as their *Ra1*, our *Nb8* as their *Pb2*; transferred move **e6d5** (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2751` (ply 22) [structural distance  $d=48$ , QAP obj=60]: the master played **Qxd2**; under  $\mu$  our *Pe3* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *pa2* as their *Ra1*, our *Ra8* as their *Ph2*; transferred move **e6d5** (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1295` (ply 21) [structural distance  $d=62$ , QAP obj=60]: the master played **dx c6**; under  $\mu$  our *Pe3* plays the role of that game’s *Pd2*; cross-type roles: our *bc1* as their *Ra8*, our *Nb8* as their *Ph7*; transferred move **e6d5** (**endorses this move**). Synthesis: 15 retrieved analogs transferred to **exd5** (summed QAP objective 667, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



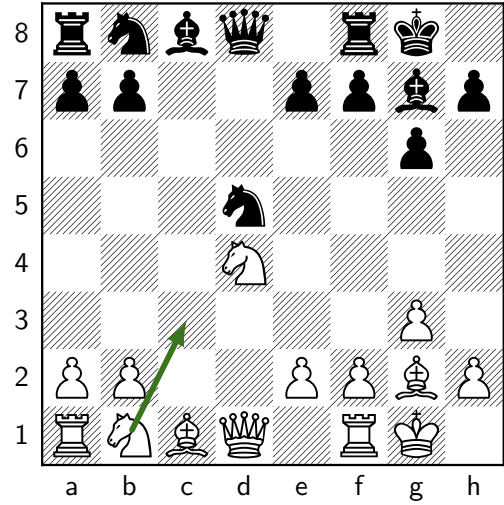
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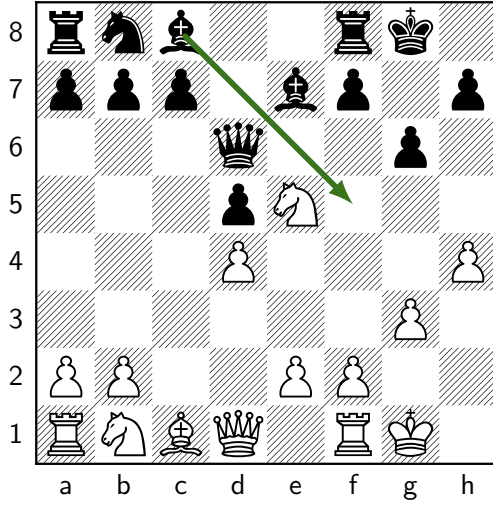


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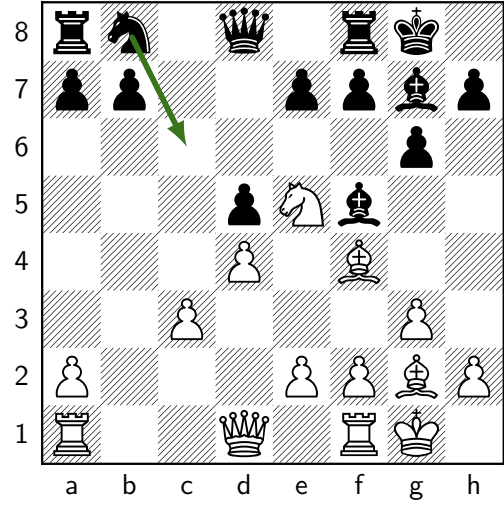


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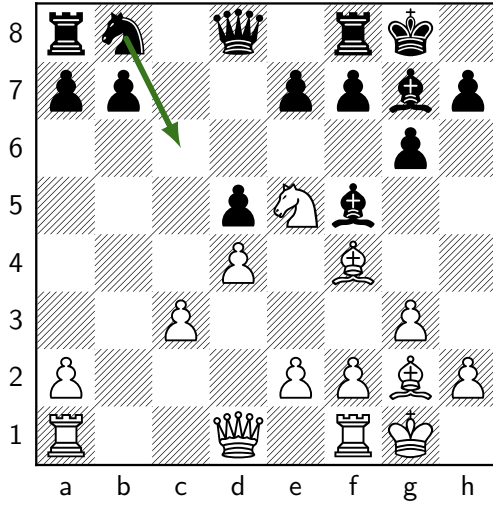
Figure 4: **Move 10, Black: Qd6.** (a) Our position; Melanie plays Qd6 (d8d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#874` (ply 16) [structural distance  $d=60$ , QAP obj=16]: the master played Nc3; under  $\mu$  our Qd1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Ra1, our Rf8 as their Qd1; transferred move d8d6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#875` (ply 16) [structural distance  $d=60$ , QAP obj=16]: the master played Nc3; under  $\mu$  our Qd1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Ra1, our Rf8 as their Qd1; transferred move d8d6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2035` (ply 16) [structural distance  $d=60$ , QAP obj=16]: the master played Nc3; under  $\mu$  our Qd1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Ra1, our Rf8 as their Qd1; transferred move d8d6 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Qd6 (summed QAP objective 291, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



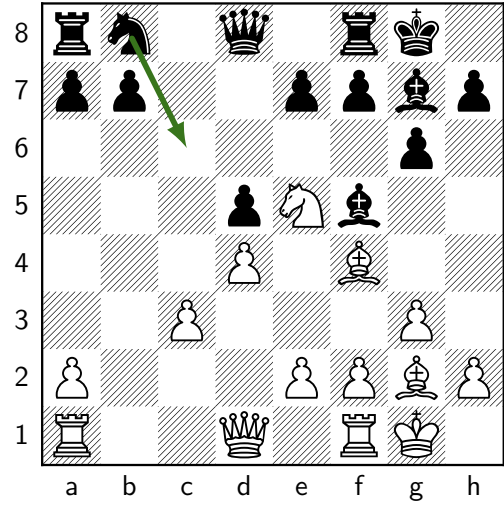
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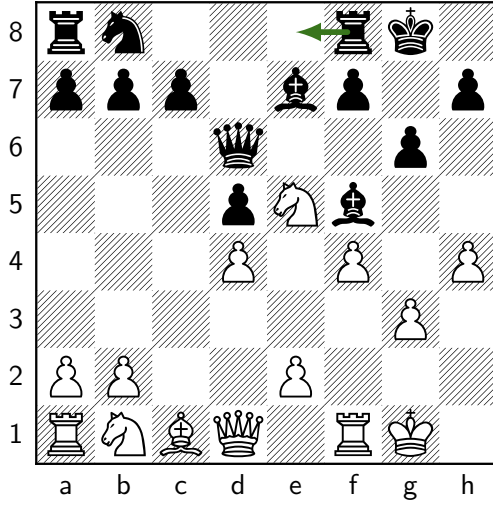


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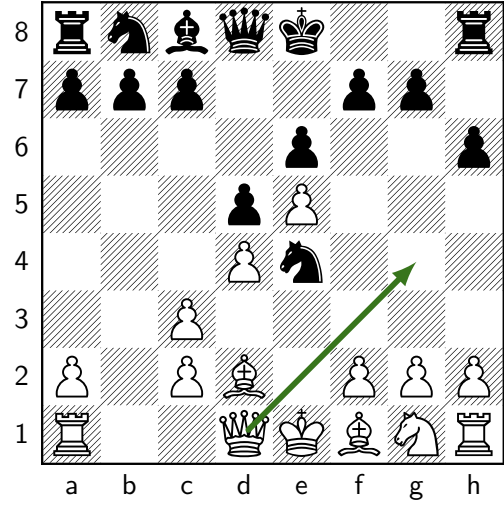


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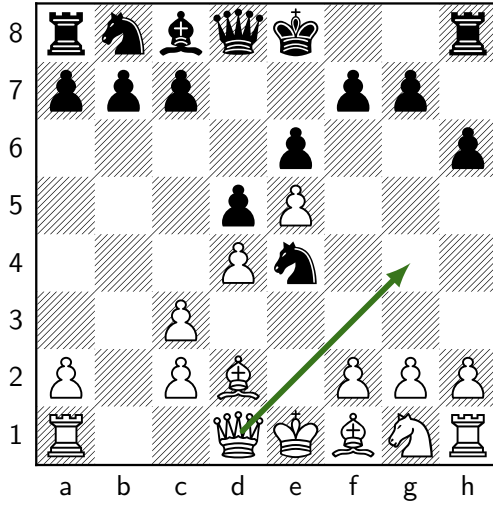
Figure 5: **Move 11, Black: Bf5**. (a) Our position; Melanie plays Bf5 (c8f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#276` (ply 21) [structural distance  $d=70$ , QAP obj=35]: the master played Nc6; under  $\mu$  our Bc1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Ra8, our rf1 as their Qd8; transferred move c8f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#277` (ply 21) [structural distance  $d=70$ , QAP obj=35]: the master played Nc6; under  $\mu$  our Bc1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Ra8, our rf1 as their Qd8; transferred move c8f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2446` (ply 21) [structural distance  $d=70$ , QAP obj=35]: the master played Nc6; under  $\mu$  our Bc1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Ra8, our rf1 as their Qd8; transferred move c8f5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bf5 (summed QAP objective 221, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



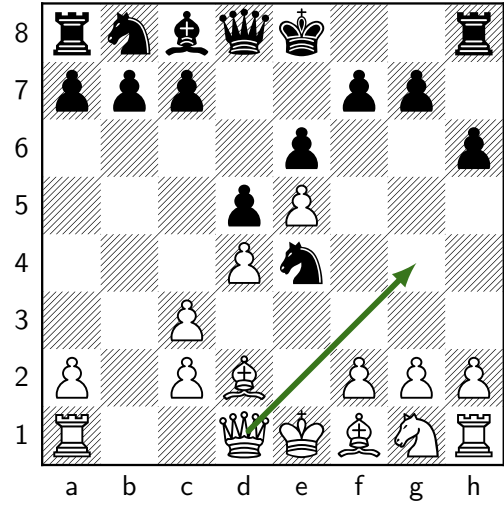
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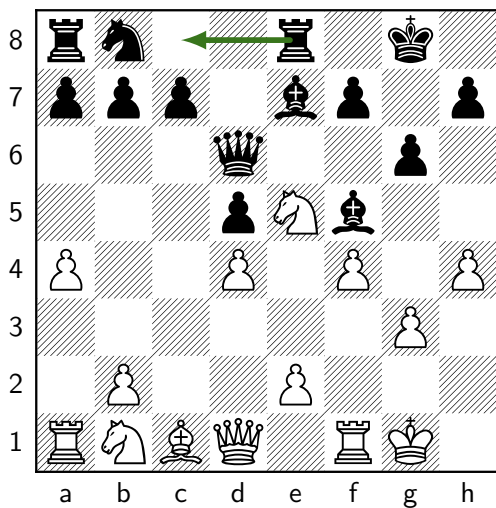


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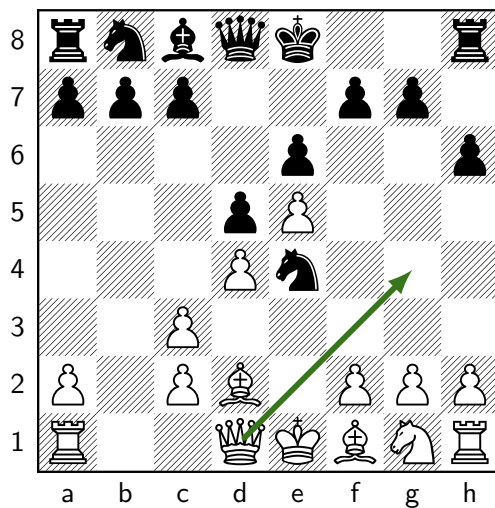


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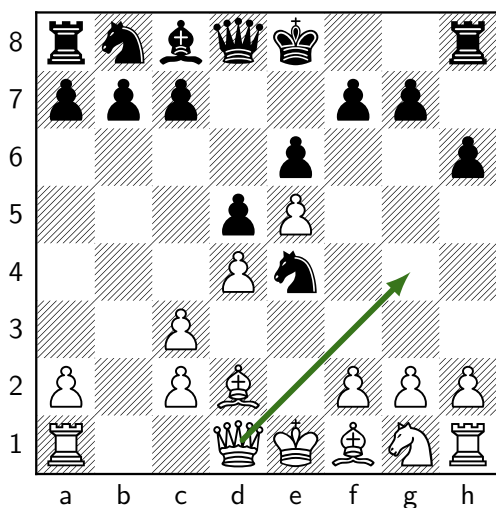
Figure 6: **Move 12, Black: Re8**. (a) Our position; Melanie plays **Re8** (**f8e8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1873` (ply 14) [structural distance  $d=75$ , QAP obj=24]: the master played **Qg4**; under  $\mu$  our *Rf1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *rf1* as their *Ng1*, our *Pa7* as their *Rh1*; transferred move **f8e8** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1874` (ply 14) [structural distance  $d=75$ , QAP obj=24]: the master played **Qg4**; under  $\mu$  our *Rf1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *rf1* as their *Ng1*, our *Pa7* as their *Rh1*; transferred move **f8e8** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1442` (ply 14) [structural distance  $d=75$ , QAP obj=24]: the master played **Qg4**; under  $\mu$  our *Rf1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *rf1* as their *Ng1*, our *Pa7* as their *Rh1*; transferred move **f8e8** (**endorses this move**). Synthesis: 8 retrieved analogs transferred to **Re8** (summed QAP objective 193, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



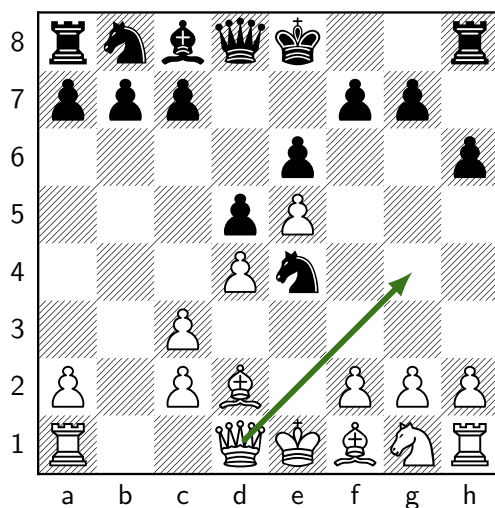
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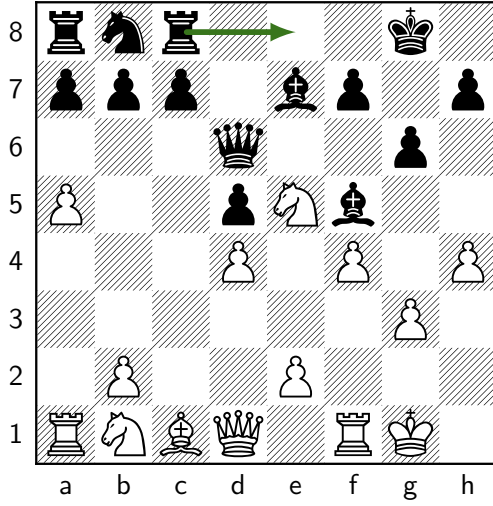
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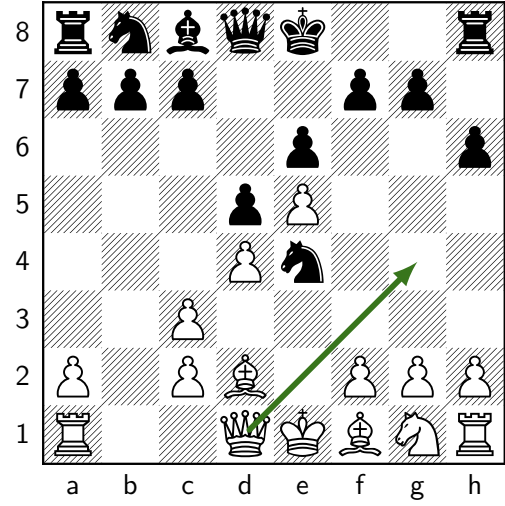
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Figure 7: **Move 13, Black: Rc8**. (a) Our position; Melanie plays Rc8 (e8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1873` (ply 14) [structural distance  $d=72$ , QAP obj=23]: the master played Qg4; under  $\mu$  our *Re1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *pf4* as their *Ra1*, our *Be7* as their *Ng1*; transferred move e8c8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1874` (ply 14) [structural distance  $d=72$ , QAP obj=23]: the master played Qg4; under  $\mu$  our *Re1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *pf4* as their *Ra1*, our *Be7* as their *Ng1*; transferred move e8c8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1442` (ply 14) [structural distance  $d=72$ , QAP obj=23]: the master played Qg4; under  $\mu$  our *Re1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *pf4* as their *Ra1*, our *Be7* as their *Ng1*; transferred move e8c8 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Rc8 (summed QAP objective 184, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

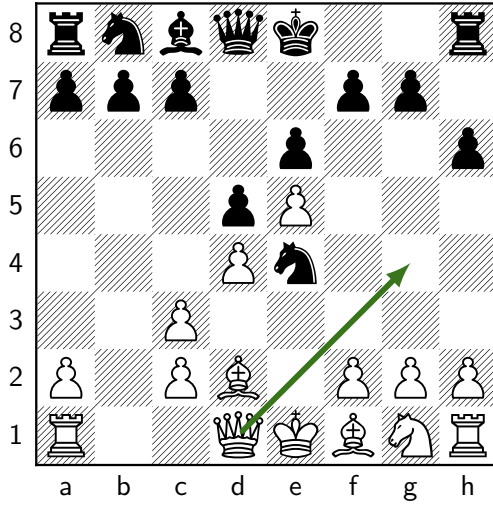




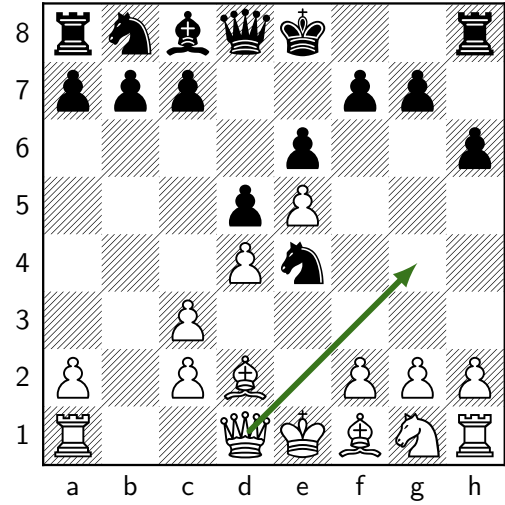
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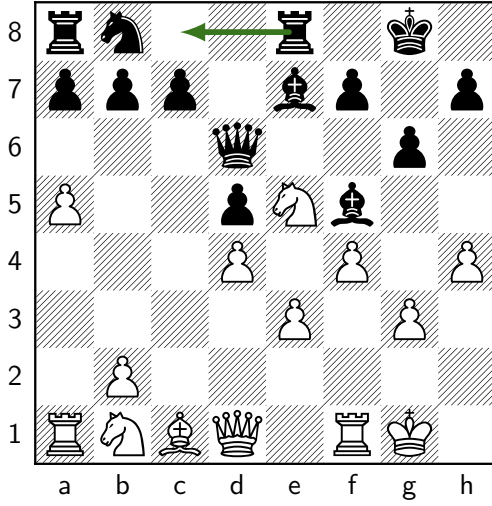


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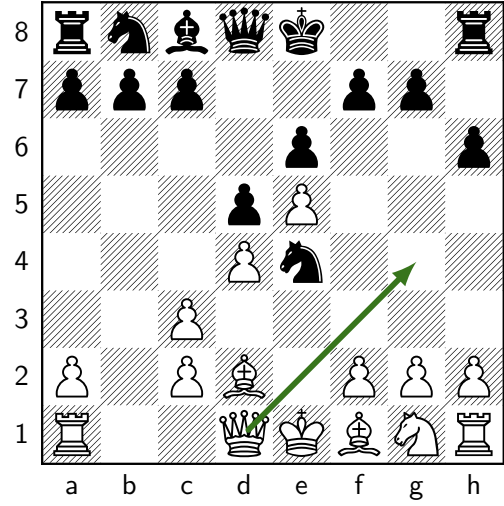


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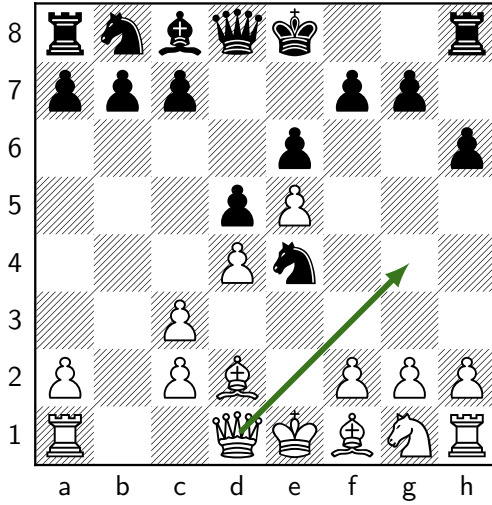
Figure 8: **Move 14, Black: Re8**. (a) Our position; Melanie plays **Re8** (**c8e8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1873` (ply 14) [structural distance  $d=81$ , QAP obj=23]: the master played **Qg4**; under  $\mu$  our *Rc1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Bf5* as their *Ra1*, our *pa5* as their *Bf1*; transferred move **c8e8** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1874` (ply 14) [structural distance  $d=81$ , QAP obj=23]: the master played **Qg4**; under  $\mu$  our *Rc1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Bf5* as their *Ra1*, our *pa5* as their *Bf1*; transferred move **c8e8** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1442` (ply 14) [structural distance  $d=81$ , QAP obj=23]: the master played **Qg4**; under  $\mu$  our *Rc1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Bf5* as their *Ra1*, our *pa5* as their *Bf1*; transferred move **c8e8** (**endorses this move**). Synthesis: 9 retrieved analogs transferred to **Re8** (summed QAP objective 186, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



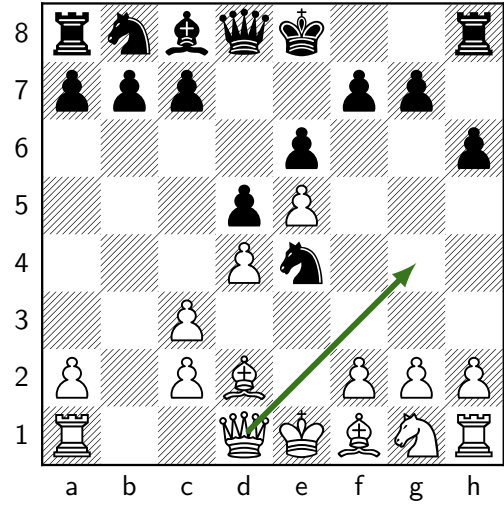
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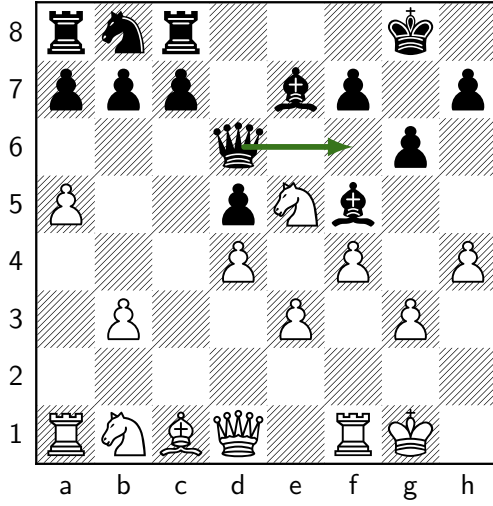


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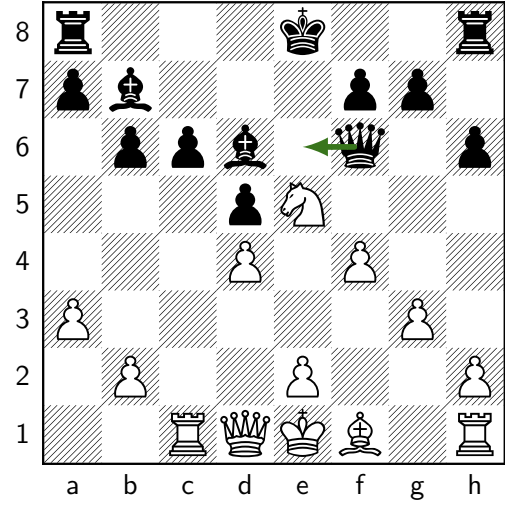


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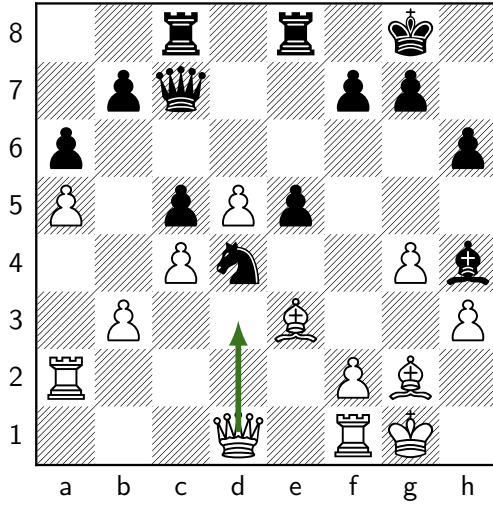
Figure 9: **Move 15, Black: Rc8**. (a) Our position; Melanie plays Rc8 (e8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1873` (ply 14) [structural distance  $d=93$ , QAP obj=24]: the master played Qg4; under  $\mu$  our *Re1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Nb8* as their *Ra1*, our *pe3* as their *Bf1*; transferred move e8c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1874` (ply 14) [structural distance  $d=93$ , QAP obj=24]: the master played Qg4; under  $\mu$  our *Re1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Nb8* as their *Ra1*, our *pe3* as their *Bf1*; transferred move e8c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1442` (ply 14) [structural distance  $d=93$ , QAP obj=24]: the master played Qg4; under  $\mu$  our *Re1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Nb8* as their *Ra1*, our *pe3* as their *Bf1*; transferred move e8c8 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Rc8 (summed QAP objective 188, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



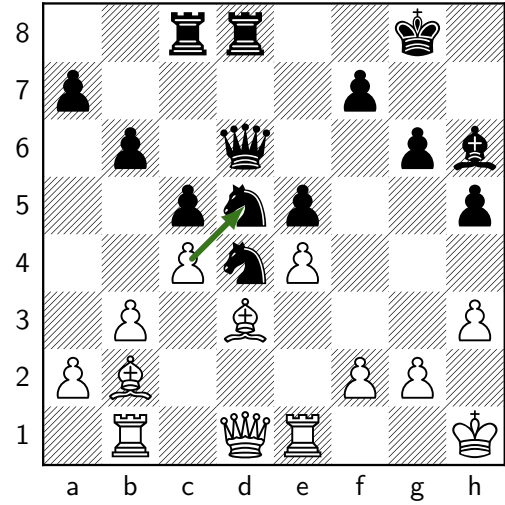
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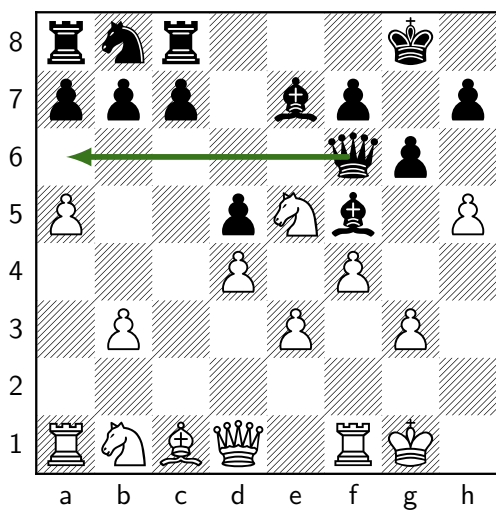


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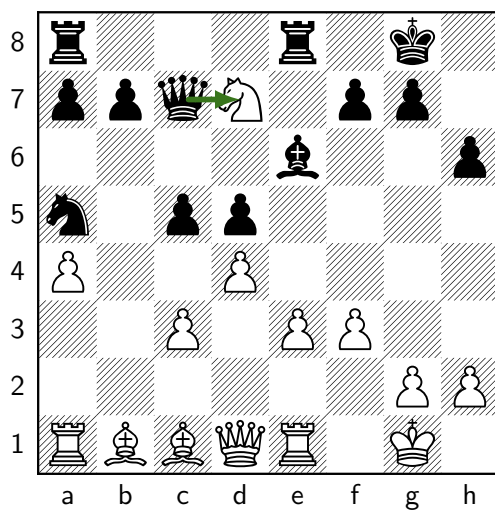


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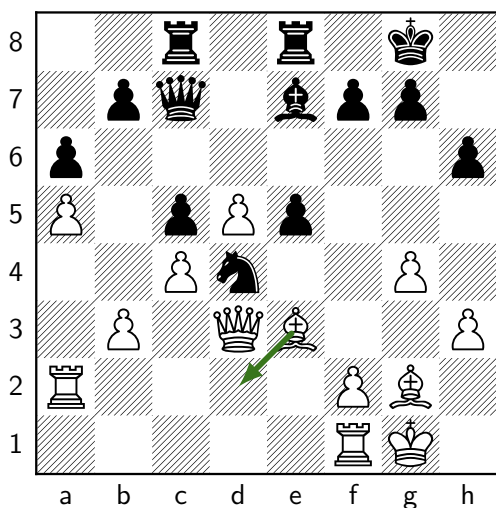
Figure 10: **Move 16, Black: Qf6**. (a) Our position; Melanie plays Qf6 (d6f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0002#565 (ply 27) [structural distance  $d=94$ , QAP obj=69]: the master played Qe6; under  $\mu$  our Qd3 plays the role of that game’s Qf3; cross-type roles: our Pc7 as their Rh8, our Ra8 as their Pa7; transferred move d6f6 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0001#3878 (ply 44) [structural distance  $d=92$ , QAP obj=56]: the master played Qd3; under  $\mu$  our Qd3 plays the role of that game’s Qd1; cross-type roles: our Be7 as their Rf1, our Bf5 as their Ra2; transferred move d6f6 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0002#1924 (ply 46) [structural distance  $d=92$ , QAP obj=43]: the master played cxd5; under  $\mu$  our Qd3 plays the role of that game’s Pc4 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Qd1, our pd4 as their Re1; transferred move d6f6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qf6 (summed QAP objective 222, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



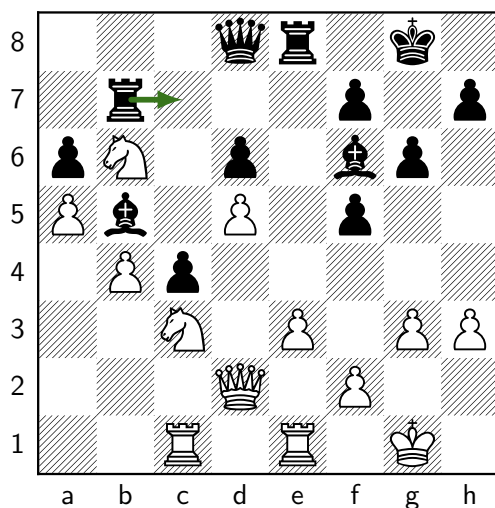
(a)



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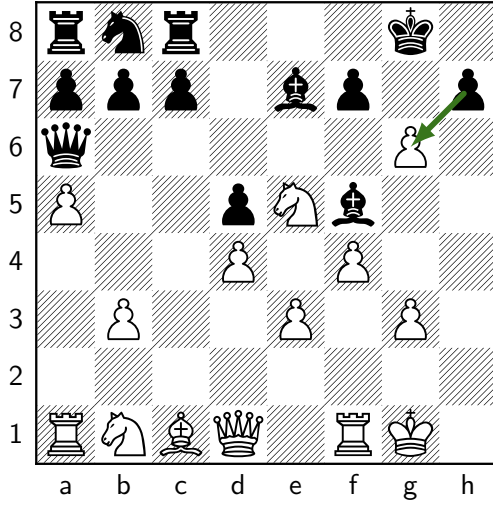


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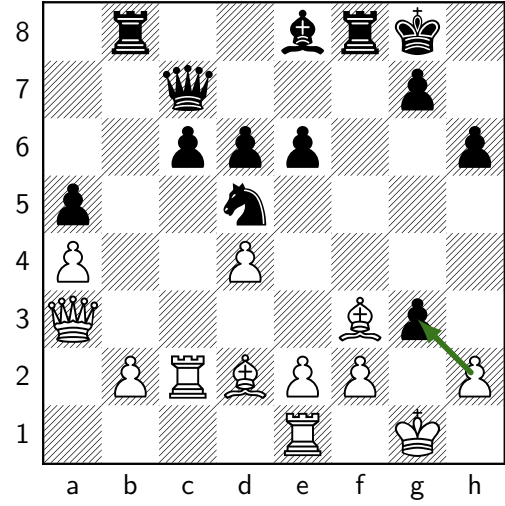


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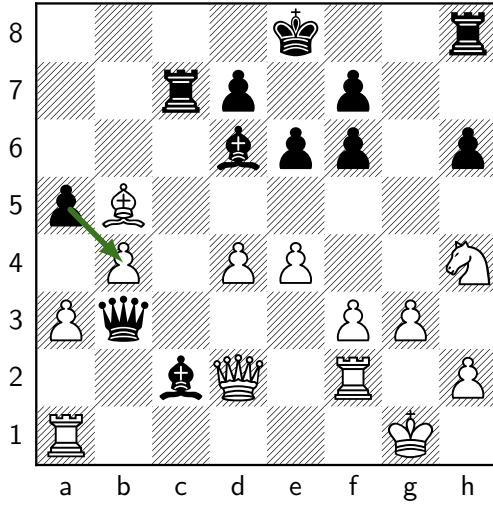
Figure 11: **Move 17, Black: Qa6**. (a) Our position; Melanie plays Qa6 (f6a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1611` (ply 31) [structural distance  $d=84$ , QAP obj=75]: the master played Qxd7; under  $\mu$  our Qf3 plays the role of that game's Qc2; cross-type roles: our Be7 as their Re8, our Bf5 as their Pa7; transferred move f6a6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3878` (ply 46) [structural distance  $d=95$ , QAP obj=55]: the master played Bd2; under  $\mu$  our Qf3 plays the role of that game's Be3 (a **cross-type** correspondence); cross-type roles: our Bf5 as their Rf1, our Nb8 as their Ra2; transferred move f6a6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1273` (ply 45) [structural distance  $d=98$ , QAP obj=54]: the master played Rc7; under  $\mu$  our Qf3 plays the role of that game's Rb2 (a **cross-type** correspondence); cross-type roles: our Be7 as their Re8, our Bf5 as their Pf7; transferred move f6a6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qa6 (summed QAP objective 252, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



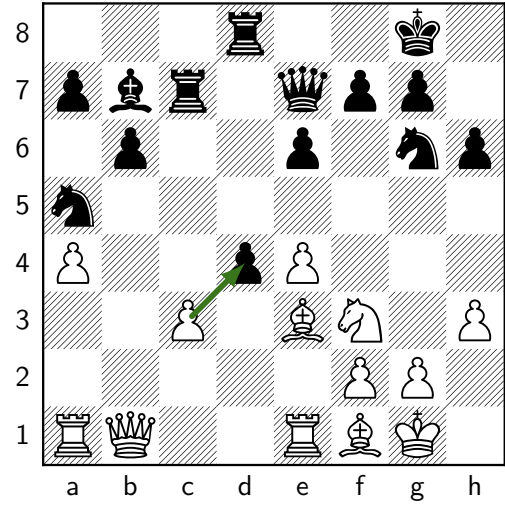
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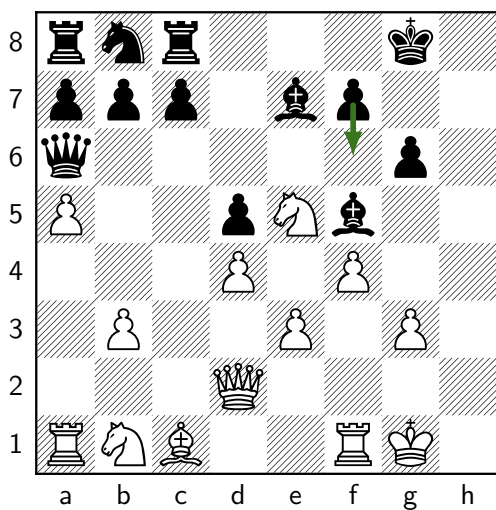


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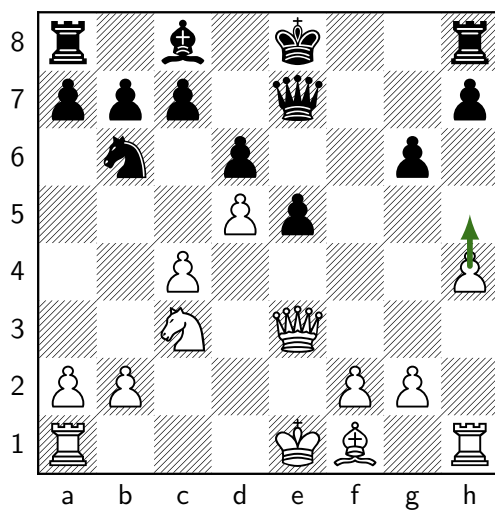


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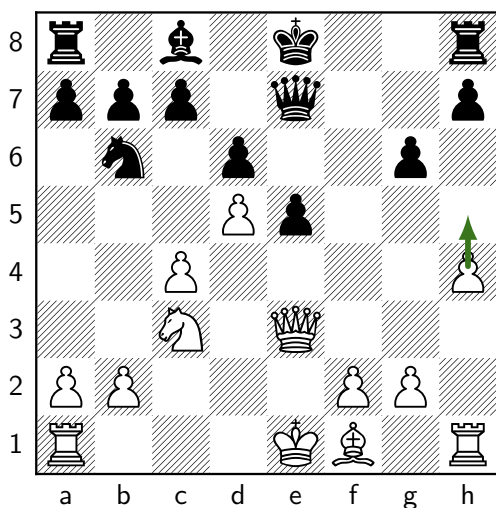
Figure 12: **Move 18, Black: hxg6.** (a) Our position; Melanie plays **hxg6** (**h7g6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2165` (ply 46) [structural distance  $d=103$ , QAP obj=109]: the master played **hxg3**; under  $\mu$  our *Ph2* plays the role of that game’s *Ph2*; cross-type roles: our *Nb8* as their *Rc2*, our *Pf7* as their *Bf3*; transferred move **h7g6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2826` (ply 49) [structural distance  $d=98$ , QAP obj=63]: the master played **axb4**; under  $\mu$  our *Ph2* plays the role of that game’s *Pa4*; cross-type roles: our *Pc7* as their *Rc7*, our *Be7* as their *Pe6*; transferred move **h7g6** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2212` (ply 44) [structural distance  $d=110$ , QAP obj=36]: the master played **cx d4**; under  $\mu$  our *Ph2* plays the role of that game’s *Pc3*; cross-type roles: our *pf4* as their *Qb1*, our *Pc7* as their *Re1*; transferred move **h7g6** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **hxg6** (summed QAP objective 208, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



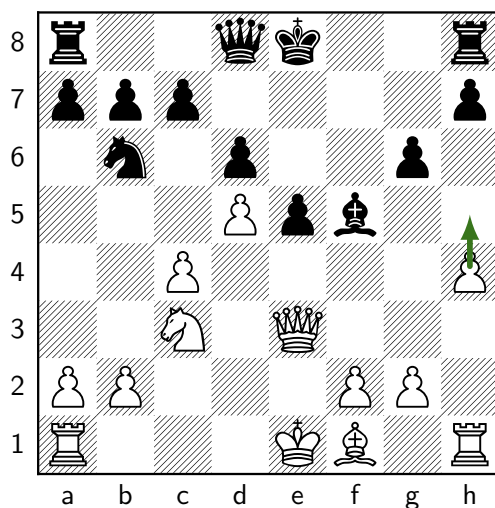
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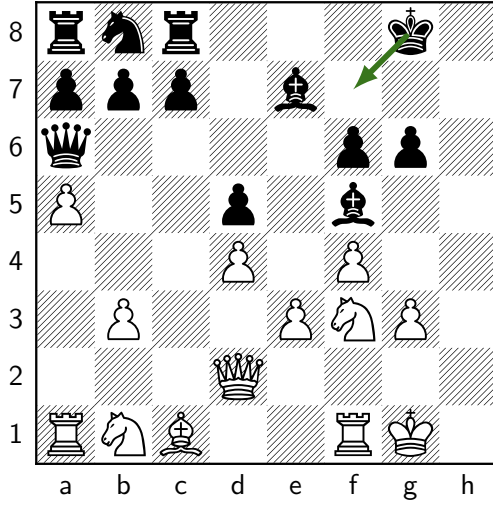


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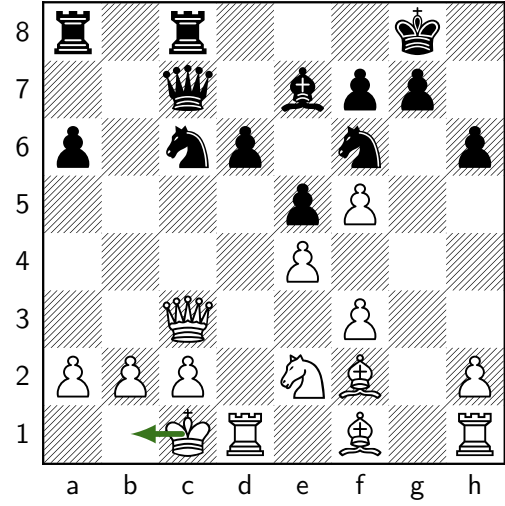


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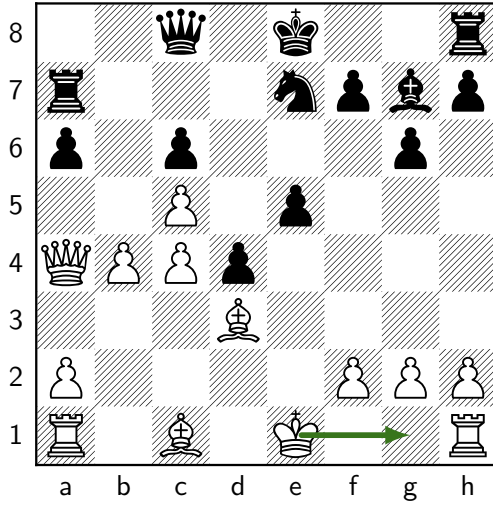
Figure 13: **Move 19, Black: f6** (f7f6). (a) Our position; Melanie plays f6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2860` (ply 28) [structural distance  $d=87$ , QAP obj=61]: the master played h5; under  $\mu$  our  $Pf2$  plays the role of that game’s  $Ph4$ ; cross-type roles: our  $pf4$  as their  $Bf1$ , our  $Be7$  as their  $Rh1$ ; transferred move f7f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2861` (ply 28) [structural distance  $d=87$ , QAP obj=61]: the master played h5; under  $\mu$  our  $Pf2$  plays the role of that game’s  $Ph4$ ; cross-type roles: our  $pf4$  as their  $Bf1$ , our  $Be7$  as their  $Rh1$ ; transferred move f7f6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#910` (ply 28) [structural distance  $d=85$ , QAP obj=59]: the master played h5; under  $\mu$  our  $Pf2$  plays the role of that game’s  $Ph4$ ; cross-type roles: our  $Pg6$  as their  $Bf1$ , our  $qd2$  as their  $Rh1$ ; transferred move f7f6 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to f6 (summed QAP objective 424, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



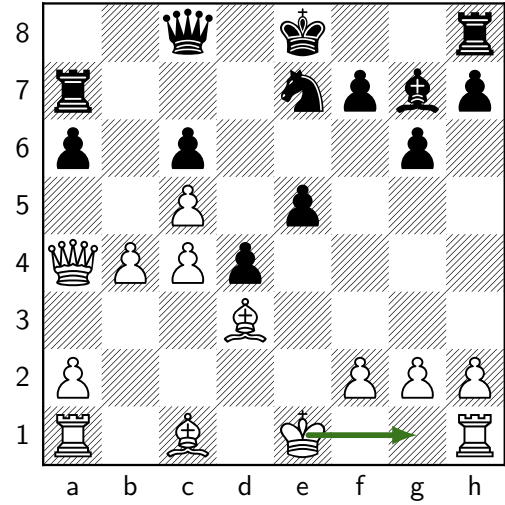
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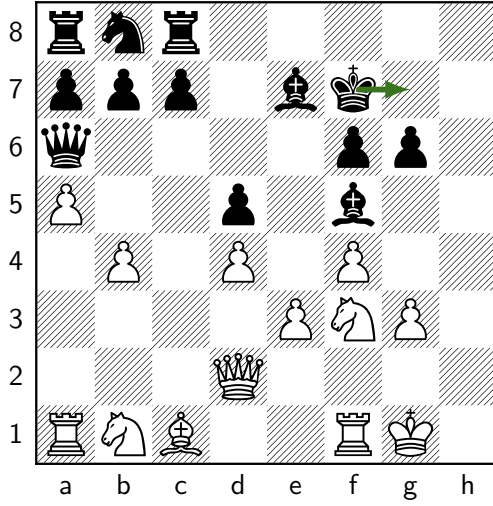
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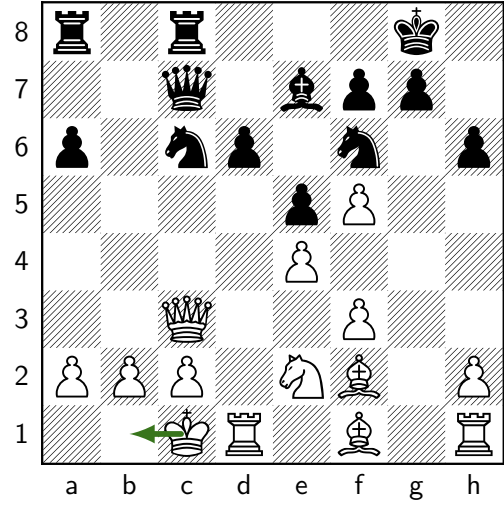
(d)

Figure 14: **Move 20, Black: Kf7**. (a) Our position; Melanie plays Kf7 (g8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1292` (ply 32) [structural distance  $d=102$ , QAP obj=75]: the master played Kb1; under  $\mu$  our Kg1 plays the role of that game’s Kc1; cross-type roles: our Pd5 as their Bf1, our Nb8 as their Rh1; transferred move g8f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2341` (ply 32) [structural distance  $d=101$ , QAP obj=56]: the master played 0-0; under  $\mu$  our Kg1 plays the role of that game’s Ke1; cross-type roles: our Nb8 as their Ra1, our Pd5 as their Bc1; transferred move g8f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2244` (ply 32) [structural distance  $d=101$ , QAP obj=56]: the master played 0-0; under  $\mu$  our Kg1 plays the role of that game’s Ke1; cross-type roles: our Nb8 as their Ra1, our Pd5 as their Bc1; transferred move g8f7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kf7 (summed QAP objective 270, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

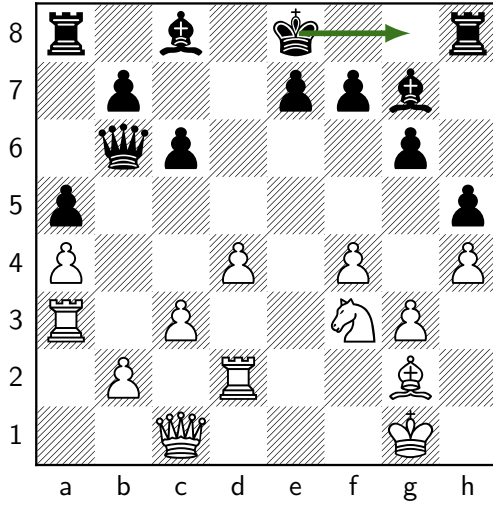




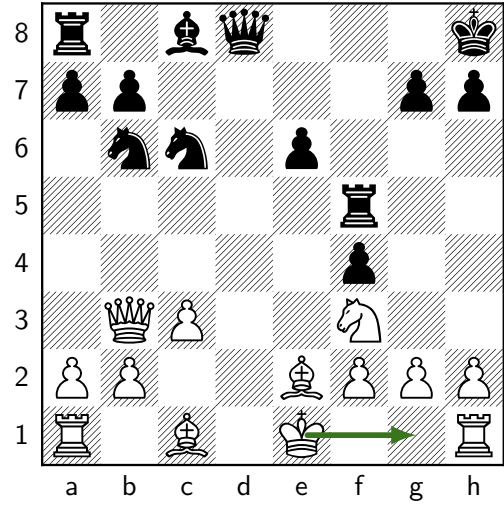
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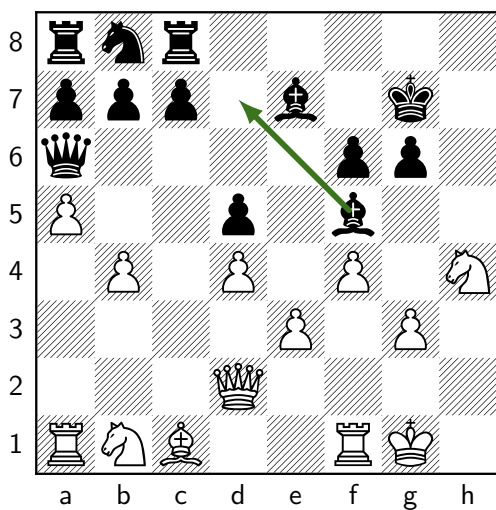
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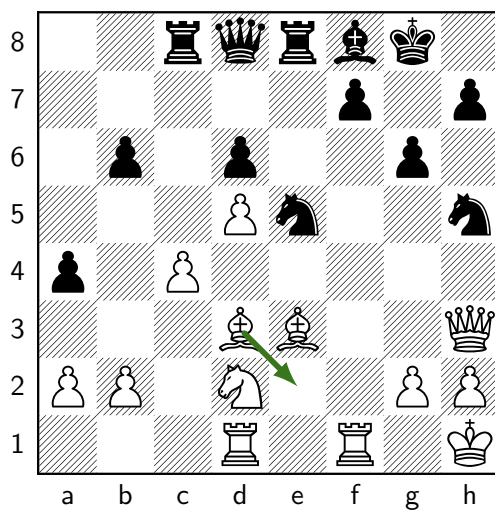
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Figure 15: **Move 21, Black: Kg7**. (a) Our position; Melanie plays Kg7 (f7g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1292` (ply 32) [structural distance  $d=96$ , QAP obj=72]: the master played Kb1; under  $\mu$  our Kf2 plays the role of that game’s Kc1; cross-type roles: our Pd5 as their Rd1, our Ra8 as their Bf1; transferred move f7g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1957` (ply 35) [structural distance  $d=95$ , QAP obj=42]: the master played 0-0; under  $\mu$  our Kf2 plays the role of that game’s Ke1; cross-type roles: our Be7 as their Ra8, our nf3 as their Bc8; transferred move f7g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2211` (ply 32) [structural distance  $d=92$ , QAP obj=37]: the master played 0-0; under  $\mu$  our Kf2 plays the role of that game’s Ke1; cross-type roles: our pf4 as their Rh1, our Pd5 as their Be2; transferred move f7g7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg7 (summed QAP objective 173, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

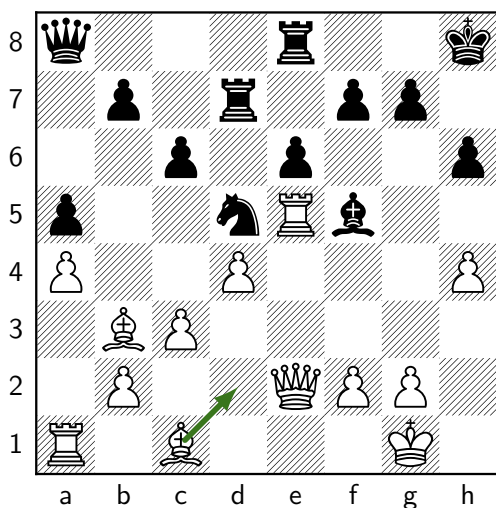




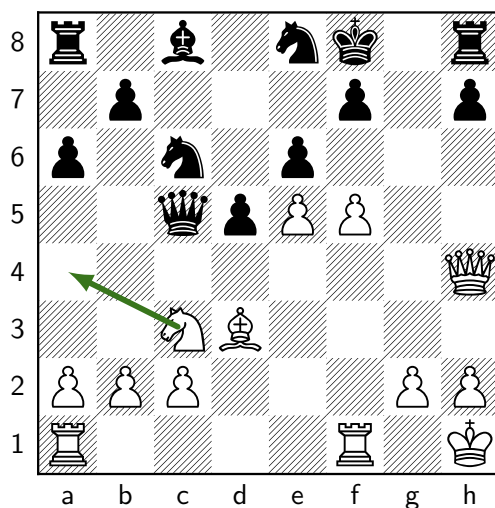
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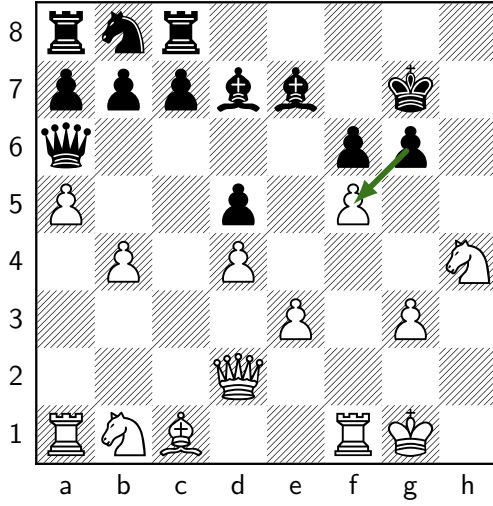


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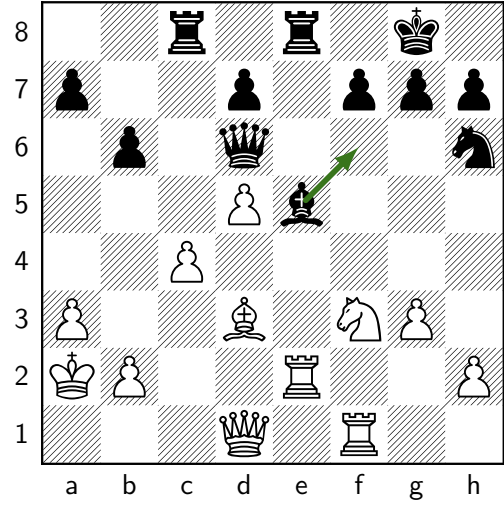


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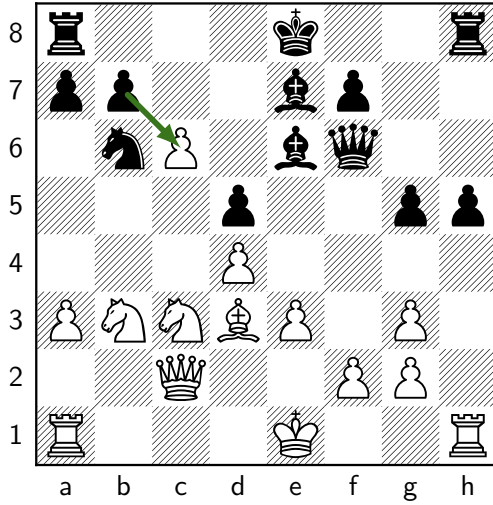
Figure 16: **Move 22, Black: Bd7.** (a) Our position; Melanie plays Bd7 (f5d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3655` (ply 42) [structural distance  $d=82$ , QAP obj=80]: the master played Be2; under  $\mu$  our Bf4 plays the role of that game’s Bd3; cross-type roles: our bc1 as their Rd1, our pe3 as their Rf1; transferred move f5d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3765` (ply 48) [structural distance  $d=82$ , QAP obj=55]: the master played Bd2; under  $\mu$  our Bf4 plays the role of that game’s Bc1; cross-type roles: our pe3 as their Ra1, our Nb8 as their Qe2; transferred move f5d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3967` (ply 36) [structural distance  $d=89$ , QAP obj=50]: the master played Na4; under  $\mu$  our Bf4 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our bc1 as their Ra1, our pg3 as their Rf1; transferred move f5d7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bd7 (summed QAP objective 322, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



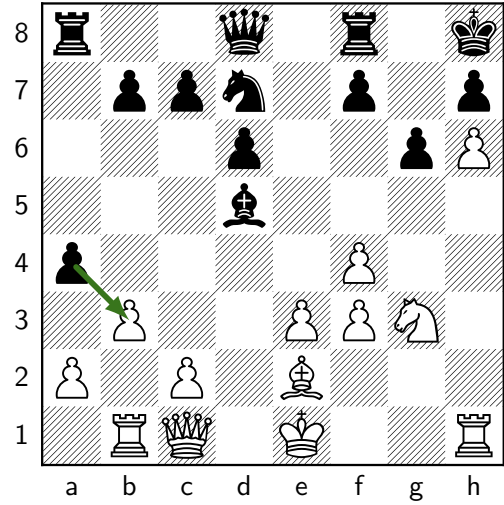
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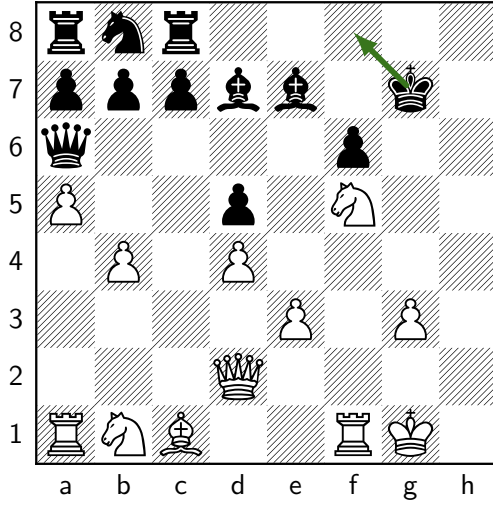


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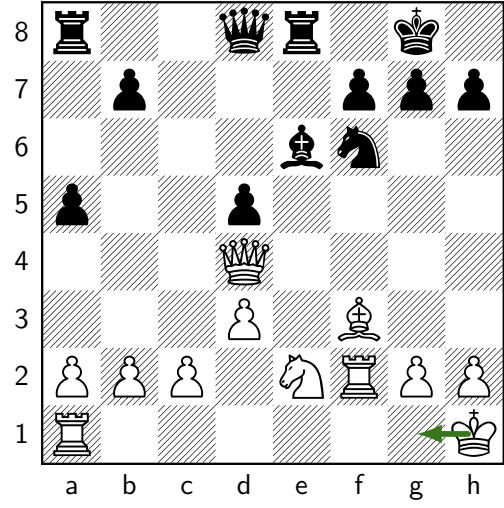


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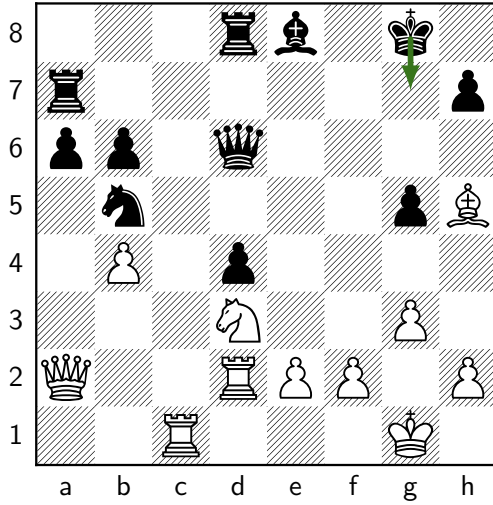
Figure 17: **Move 23, Black: gxf5.** (a) Our position; Melanie plays gxf5 (g6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1744` (ply 61) [structural distance  $d=92$ , QAP obj=84]: the master played Bf6; under  $\mu$  our Pg3 plays the role of that game’s Be4 (a **cross-type** correspondence); cross-type roles: our Pb7 as their Rc8, our Pf6 as their Re8; transferred move g6f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#235` (ply 35) [structural distance  $d=87$ , QAP obj=79]: the master played bxc6; under  $\mu$  our Pg3 plays the role of that game’s Pb2; cross-type roles: our Bd7 as their Ra8, our Pf6 as their Rh8; transferred move g6f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1181` (ply 37) [structural distance  $d=89$ , QAP obj=59]: the master played axb3; under  $\mu$  our Pg3 plays the role of that game’s Pa5; cross-type roles: our Nb8 as their Ra8, our Pf6 as their Qd8; transferred move g6f5 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to gxf5 (summed QAP objective 407, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



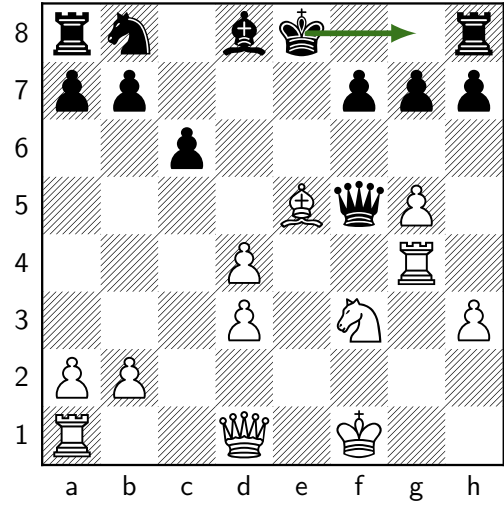
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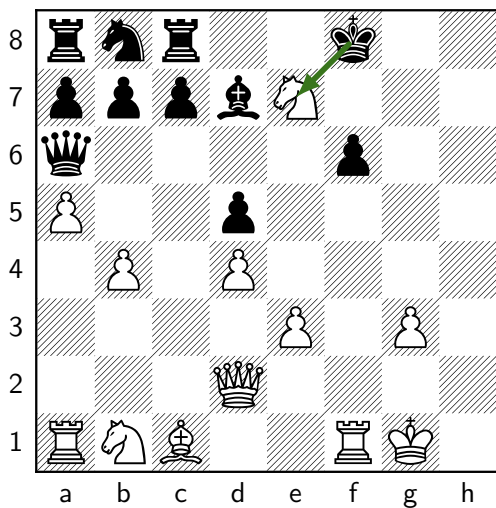


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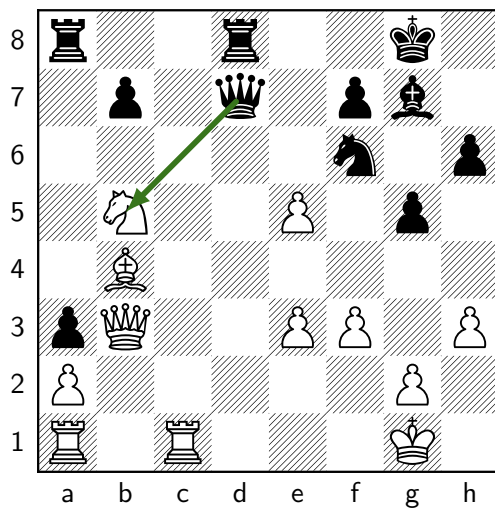


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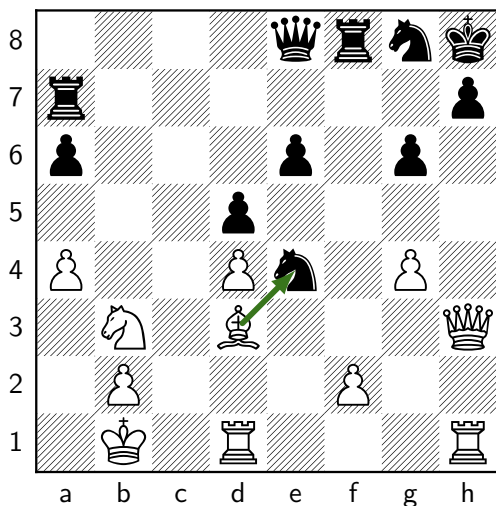
Figure 18: **Move 24, Black: Kf8.** (a) Our position; Melanie plays Kf8 (g7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2893` (ply 38) [structural distance  $d=90$ , QAP obj=33]: the master played Kg1; under  $\mu$  our Kg2 plays the role of that game’s Kh1; cross-type roles: our Pd5 as their Ra1, our Ra8 as their Pa2; transferred move g7f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3289` (ply 59) [structural distance  $d=92$ , QAP obj=27]: the master played Kg7; under  $\mu$  our Kg2 plays the role of that game’s Kg1; cross-type roles: our bc1 as their Rd8, our pb4 as their Be8; transferred move g7h8. (d) Retrieved from `ccr1_top_p0000#424` (ply 37) [structural distance  $d=94$ , QAP obj=16]: the master played O-O; under  $\mu$  our Kg2 plays the role of that game’s Ke1; cross-type roles: our pb4 as their Nb8, our Pf6 as their Bd8; transferred move g7f7. Synthesis: 1 retrieved analog transferred to Kf8 (summed QAP objective 33, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



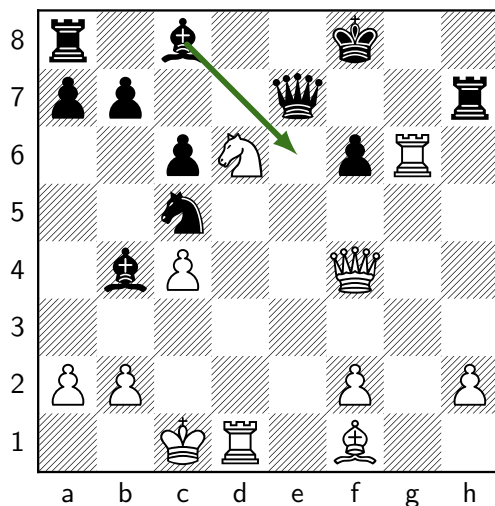
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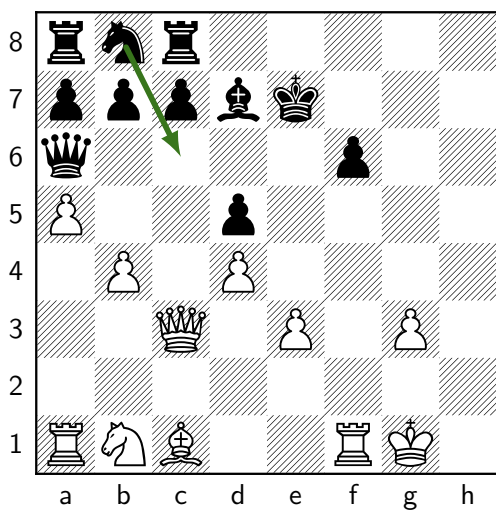


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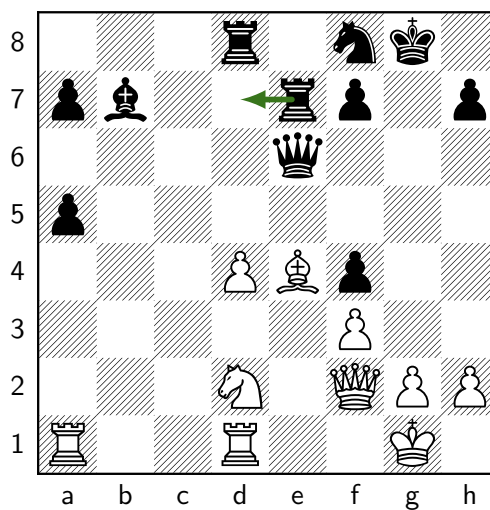


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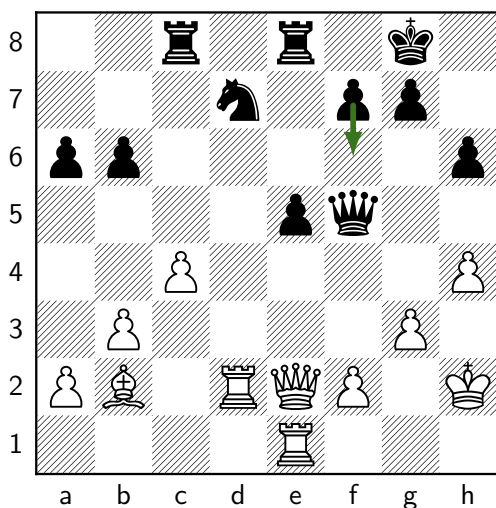
Figure 19: **Move 25, Black: Kxe7**. (a) Our position; Melanie plays Kxe7 (f8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#422` (ply 57) [structural distance  $d=85$ , QAP obj=35]: the master played Qxb5; under  $\mu$  our Kf1 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our pb4 as their Rd8, our Nb8 as their Pb7; transferred move f8e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1940` (ply 62) [structural distance  $d=85$ , QAP obj=28]: the master played Bxe4; under  $\mu$  our Kf1 plays the role of that game’s Bd3 (a **cross-type** correspondence); cross-type roles: our bc1 as their Rd1, our Bd7 as their Pb2; transferred move f8e7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#885` (ply 43) [structural distance  $d=85$ , QAP obj=56]: the master played Be6; under  $\mu$  our Rc1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Pa7, our Kf8 as their Qe7; transferred move c8d8. Synthesis: 2 retrieved analogs transferred to Kxe7 (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



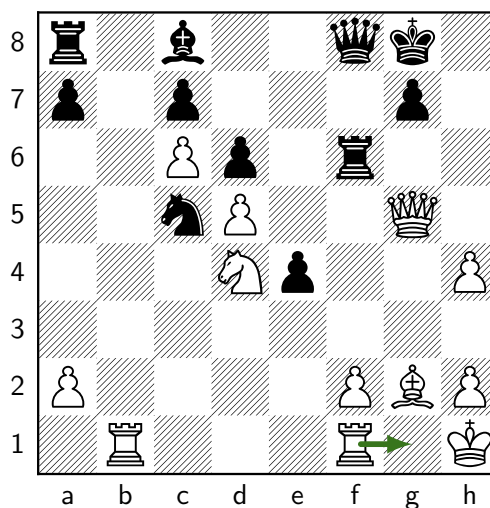
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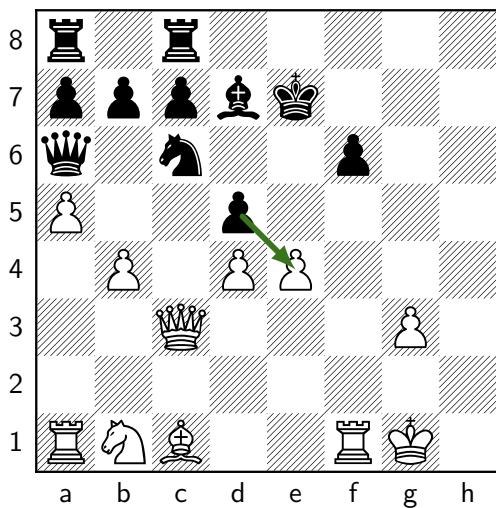


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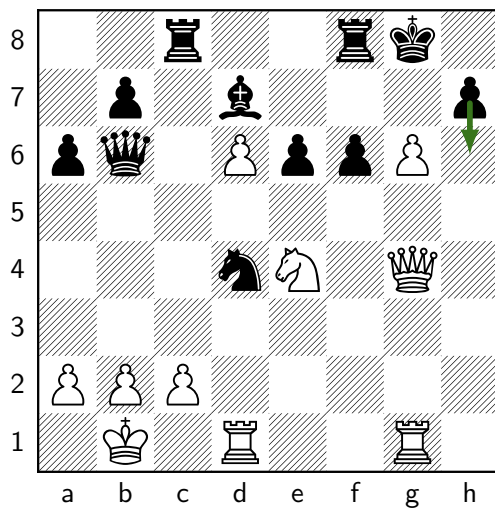


(d)

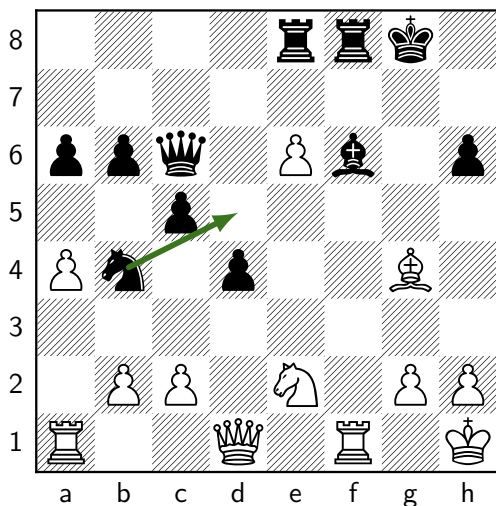
Figure 20: **Move 26, Black: Nc6**. (a) Our position; Melanie plays Nc6 (b8c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2473` (ply 51) [structural distance  $d=77$ , QAP obj=33]: the master played Red7; under  $\mu$  our Nb1 plays the role of that game’s Re2 (a **cross-type** correspondence); cross-type roles: our qc3 as their Rd8, our pe3 as their Nf8; transferred move b8c6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3561` (ply 65) [structural distance  $d=77$ , QAP obj=31]: the master played f6; under  $\mu$  our Nb1 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our pb4 as their Rc8, our rf1 as their Nd7; transferred move b8c6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1063` (ply 50) [structural distance  $d=75$ , QAP obj=25]: the master played Rg1; under  $\mu$  our Nb1 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Rb1, our Ra8 as their Pa2; transferred move b8c6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nc6 (summed QAP objective 114, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



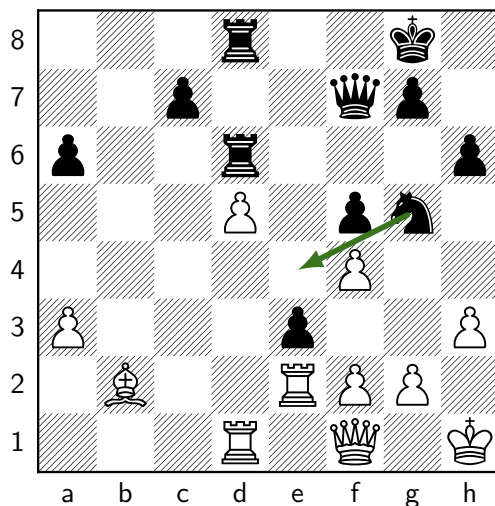
(a)



(b)

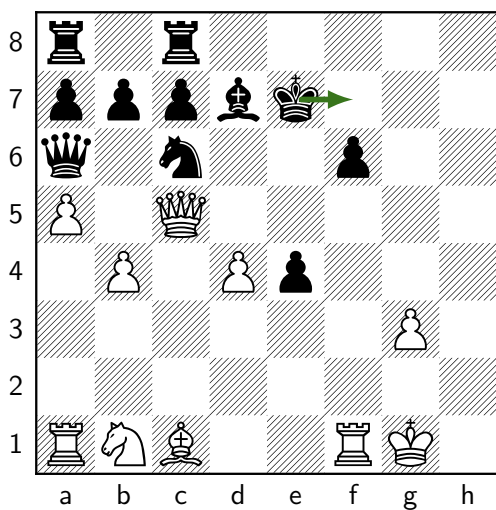


(c)

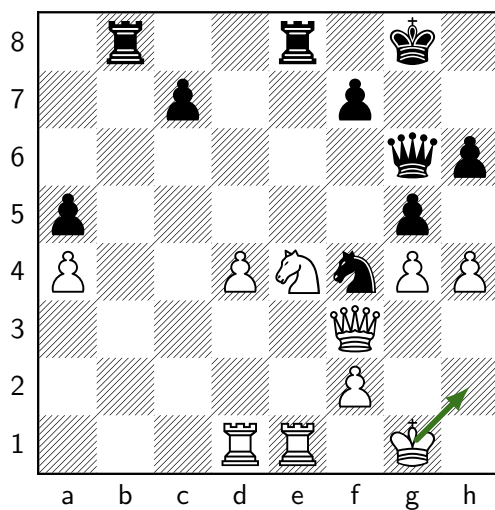


(d)

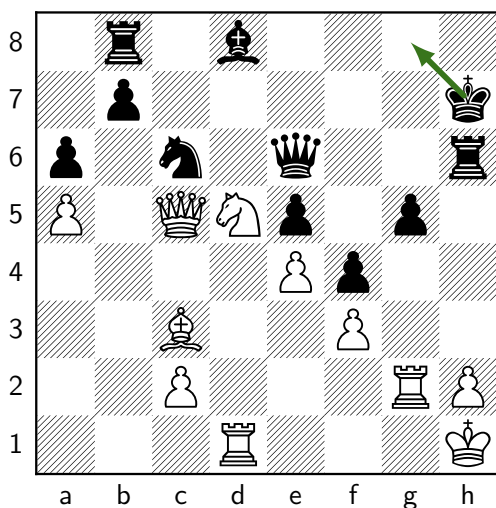
Figure 21: **Move 27, Black: dxe4**. (a) Our position; Melanie plays dxe4 (d5e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2249` (ply 53) [structural distance  $d=86$ , QAP obj=112]: the master played h6; under  $\mu$  our  $Pd4$  plays the role of that game’s  $Ph2$ ; cross-type roles: our  $Bd7$  as their  $Rc8$ , our  $Pf6$  as their  $Rf8$ ; transferred move d5e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1780` (ply 53) [structural distance  $d=83$ , QAP obj=81]: the master played Nd5; under  $\mu$  our  $Pd4$  plays the role of that game’s  $Nb5$  (a **cross-type** correspondence); cross-type roles: our  $Qa6$  as their  $Re8$ , our  $Pa7$  as their  $Rf8$ ; transferred move d5e4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1005` (ply 67) [structural distance  $d=88$ , QAP obj=66]: the master played Ne4; under  $\mu$  our  $Pd4$  plays the role of that game’s  $Ng4$  (a **cross-type** correspondence); cross-type roles: our  $Pa7$  as their  $Rd8$ , our  $Nc6$  as their  $Qf7$ ; transferred move d5e4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to dxe4 (summed QAP objective 326, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



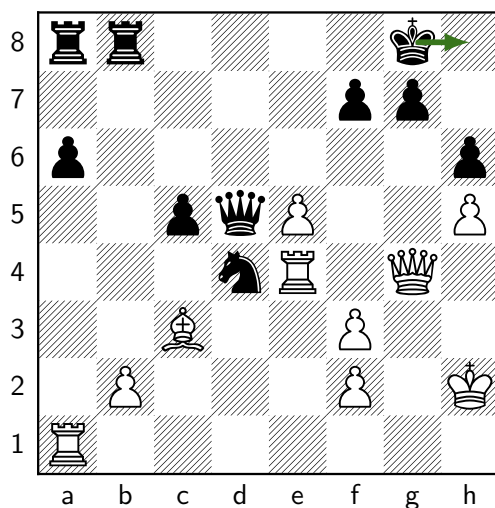
(a)



(b)



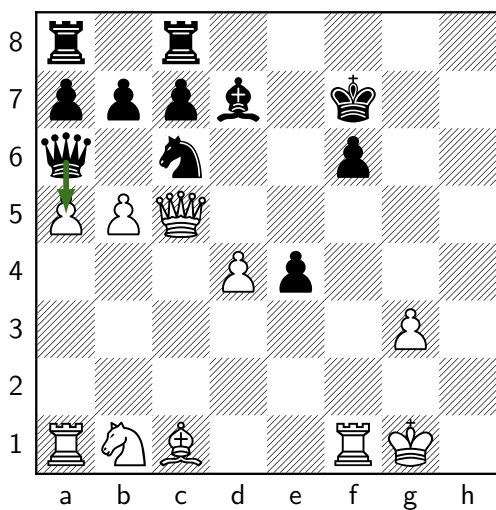
(c)



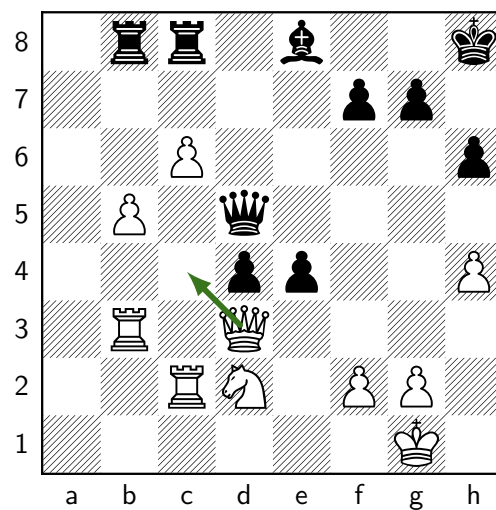
(d)

Figure 22: **Move 28, Black: Kf7**. (a) Our position; Melanie plays Kf7 (e7f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1059` (ply 60) [structural distance  $d=104$ , QAP obj=34]: the master played Kh2; under  $\mu$  our Ke2 plays the role of that game’s Kg1; cross-type roles: our Nc6 as their Re1, our ra1 as their Qf3; transferred move e7f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#300` (ply 69) [structural distance  $d=100$ , QAP obj=19]: the master played Kg8; under  $\mu$  our Ke2 plays the role of that game’s Kh2; cross-type roles: our rf1 as their Bd8, our bc1 as their Pb7; transferred move e7f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2343` (ply 51) [structural distance  $d=102$ , QAP obj=18]: the master played Kh8; under  $\mu$  our Ke2 plays the role of that game’s Kg1; cross-type roles: our pd4 as their Ra8, our nb1 as their Rb8; transferred move e7f7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kf7 (summed QAP objective 81, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

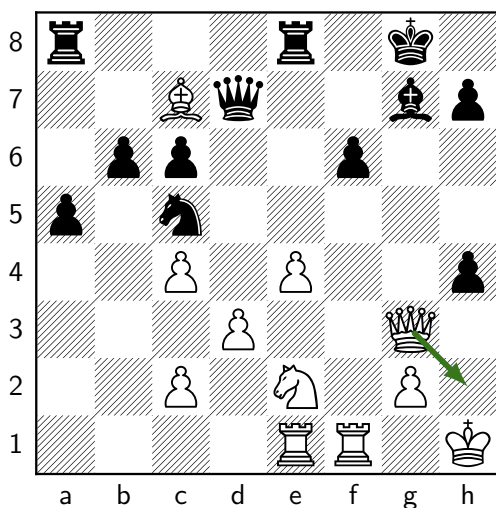




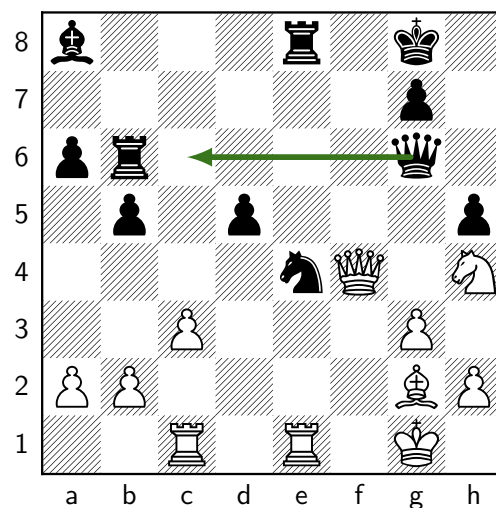
(a)



(b)

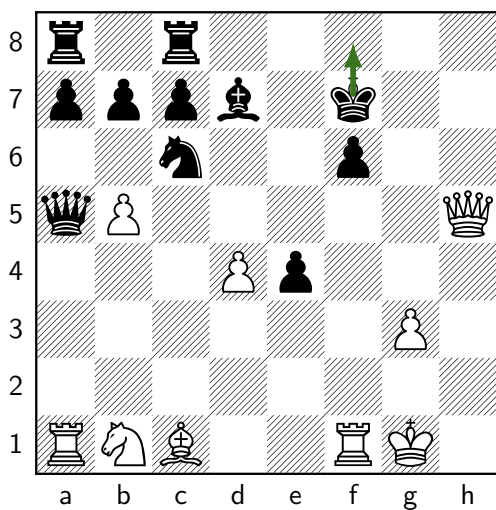


(c)

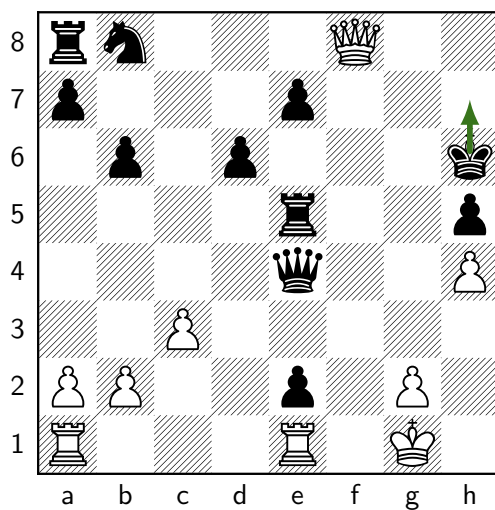


(d)

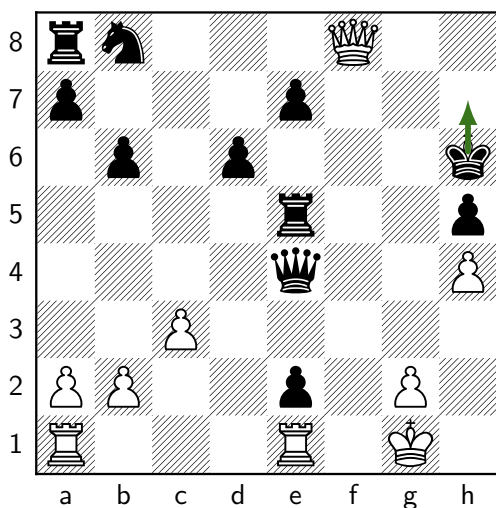
Figure 23: **Move 29, Black: Qxa5.** (a) Our position; Melanie plays Qxa5 (a6a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1102` (ply 68) [structural distance  $d=111$ , QAP obj=94]: the master played Qc4; under  $\mu$  our Qa3 plays the role of that game’s Qd3; cross-type roles: our Pa7 as their Rc2, our Ra8 as their Pg2; transferred move a6a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#931` (ply 50) [structural distance  $d=102$ , QAP obj=66]: the master played Qh2; under  $\mu$  our Qa3 plays the role of that game’s Qg3; cross-type roles: our Pe4 as their Re1, our Ra8 as their Pc2; transferred move a6a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1641` (ply 67) [structural distance  $d=114$ , QAP obj=61]: the master played Qc6; under  $\mu$  our Qa3 plays the role of that game’s Qg3; cross-type roles: our Pe4 as their Ba8, our pg3 as their Re8; transferred move a6a5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Qxa5 (summed QAP objective 348, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



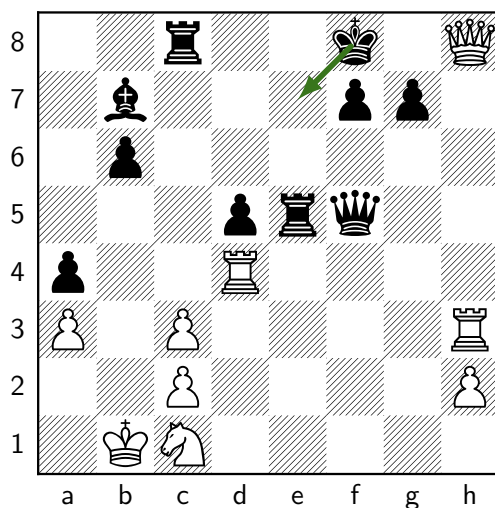
(a)



(b)

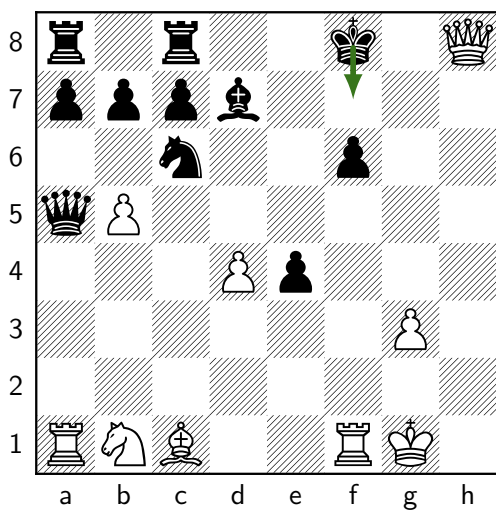


(c)

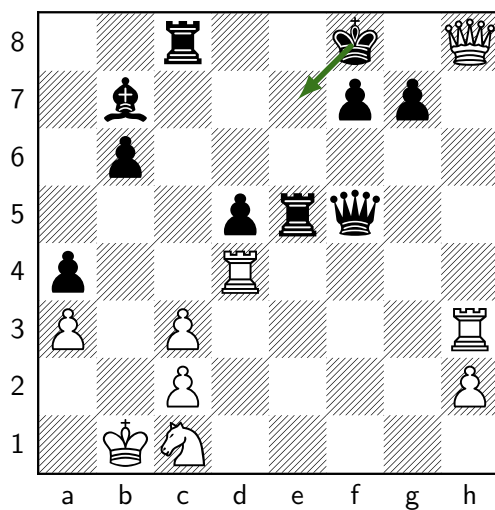


(d)

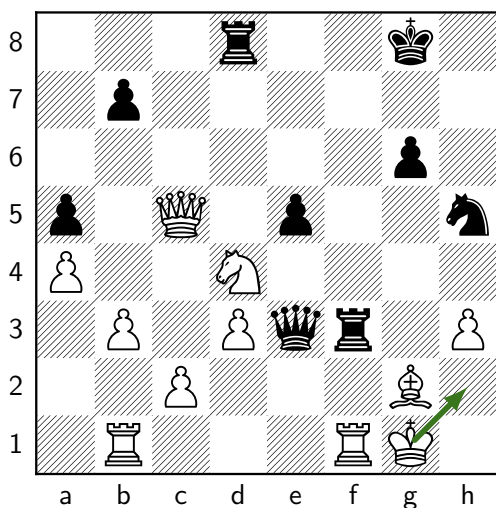
Figure 24: **Move 30, Black: Kf8 (f7f8).** (a) Our position; Melanie plays Kf8 (f7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2676` (ply 53) [structural distance  $d=83$ , QAP obj=86]: the master played Kh7; under  $\mu$  our Kf2 plays the role of that game’s Kh3; cross-type roles: our Rc8 as their Nb8, our Nc6 as their Pb6; transferred move f7f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2695` (ply 53) [structural distance  $d=83$ , QAP obj=86]: the master played Kh7; under  $\mu$  our Kf2 plays the role of that game’s Kh3; cross-type roles: our Rc8 as their Nb8, our Nc6 as their Pb6; transferred move f7f8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3146` (ply 61) [structural distance  $d=81$ , QAP obj=76]: the master played Ke7; under  $\mu$  our Kf2 plays the role of that game’s Kf1; cross-type roles: our Pa7 as their Rc8, our Ra8 as their Bb7; transferred move f7g8. Synthesis: 4 retrieved analogs transferred to Kf8 (summed QAP objective 332, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



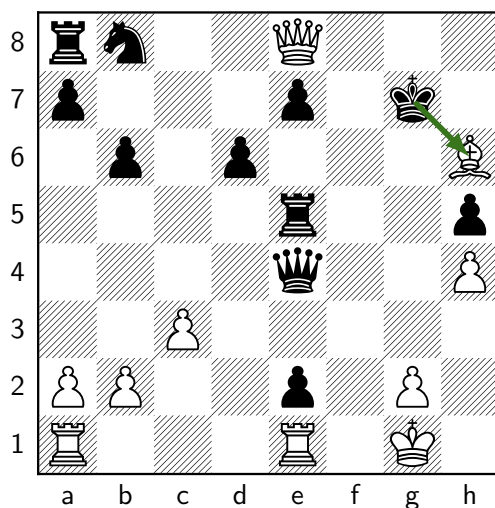
(a)



(b)

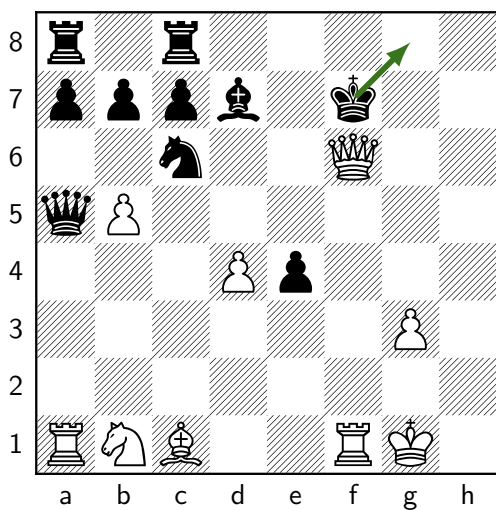


(c)

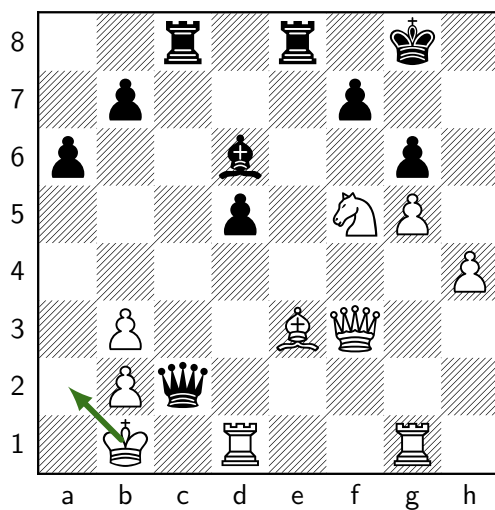


(d)

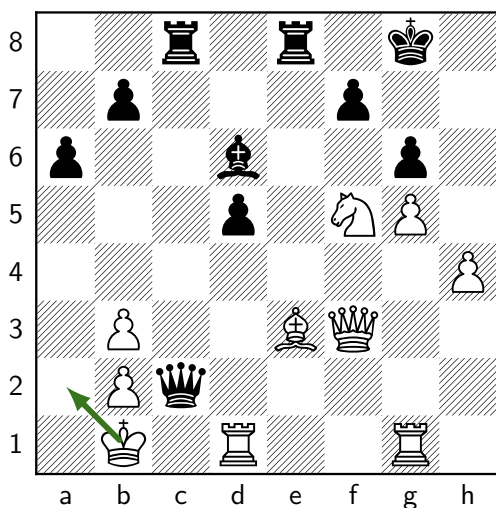
Figure 25: **Move 31, Black: Kf7 (f8f7).** (a) Our position; Melanie plays Kf7 (f8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3146` (ply 61) [structural distance  $d=93$ , QAP obj=70]: the master played Ke7; under  $\mu$  our Kf1 plays the role of that game's Kf1; cross-type roles: our Ra8 as their Bb7, our bc1 as their Pf7; transferred move f8f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#626` (ply 64) [structural distance  $d=96$ , QAP obj=69]: the master played Kh2; under  $\mu$  our Kf1 plays the role of that game's Kg1; cross-type roles: our Bd7 as their Pc2, our Pf6 as their Bg2; transferred move f8f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2676` (ply 51) [structural distance  $d=87$ , QAP obj=36]: the master played Kxh6; under  $\mu$  our Kf1 plays the role of that game's Kg2; cross-type roles: our Pc7 as their Nb8, our pb5 as their qe8; transferred move f8f7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kf7 (summed QAP objective 279, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



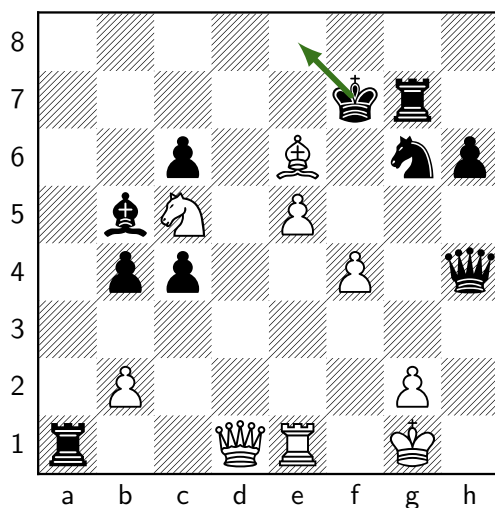
(a)



(b)

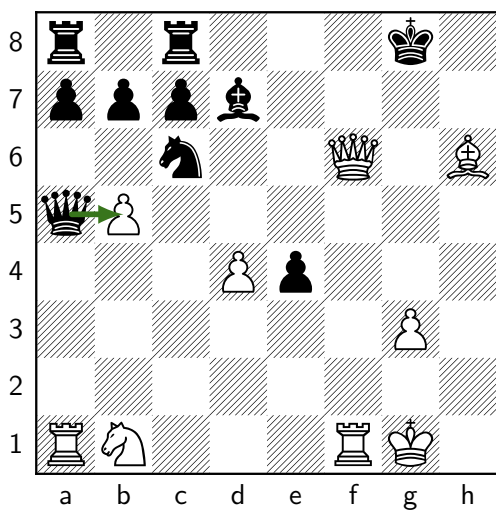


(c)

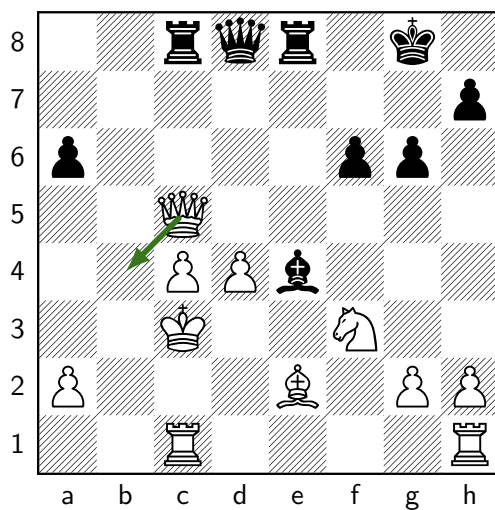


(d)

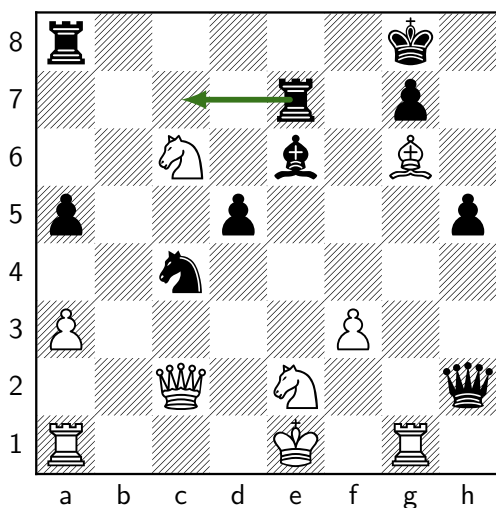
Figure 26: **Move 32, Black: Kg8.** (a) Our position; Melanie plays Kg8 (f7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#172` (ply 46) [structural distance  $d=89$ , QAP obj=74]: the master played Ka2; under  $\mu$  our Kf2 plays the role of that game’s Kb1; cross-type roles: our Pc7 as their Be3, our Bd7 as their Qf3; transferred move f7g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1869` (ply 46) [structural distance  $d=89$ , QAP obj=74]: the master played Ka2; under  $\mu$  our Kf2 plays the role of that game’s Kb1; cross-type roles: our Pc7 as their Be3, our Bd7 as their Qf3; transferred move f7g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3012` (ply 61) [structural distance  $d=98$ , QAP obj=68]: the master played Ke8; under  $\mu$  our Kf2 plays the role of that game’s Kf2; cross-type roles: our qf6 as their be6, our Pe4 as their Ng6; transferred move f7g8 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kg8 (summed QAP objective 376, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



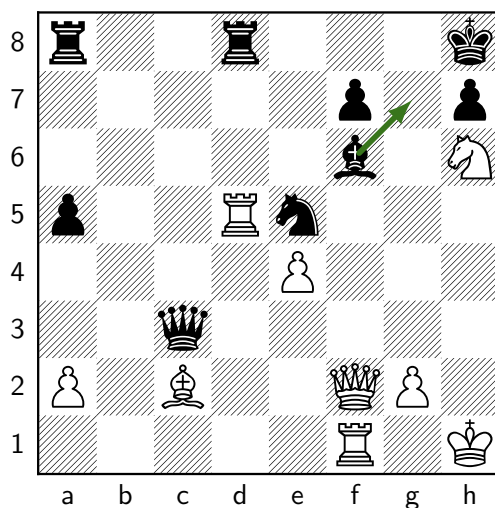
(a)



(b)



(c)



(d)

Figure 27: **Move 33, Black: Qxb5.** (a) Our position; Melanie plays Qxb5 (a5b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#767` (ply 52) [structural distance  $d=89$ , QAP obj=42]: the master played Qb4; under  $\mu$  our Qa4 plays the role of that game's Qc5; cross-type roles: our Pc7 as their Rh1, our Bd7 as their Pa2; transferred move a5b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3854` (ply 57) [structural distance  $d=97$ , QAP obj=21]: the master played Rc7; under  $\mu$  our Qa4 plays the role of that game's Re2 (a **cross-type** correspondence); cross-type roles: our qf6 as their Ra8, our ra1 as their nc6; transferred move a5b5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2959` (ply 63) [structural distance  $d=91$ , QAP obj=50]: the master played Bg7; under  $\mu$  our Nc3 plays the role of that game's Bf3 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Ra8, our Qa5 as their Rd8; transferred move c6b8. Synthesis: 5 retrieved analogs transferred to Qxb5 (summed QAP objective 168, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.